ABSTRACT

THE PERCEPTION OF PORTUGUESE VOWELS BY BILINGUAL AND HERITAGE CALIFORNIAN ENGLISH LEARNERS OF PORTUGUESE

The present study investigated the influence of language background (bilingual & heritage) and dialect (Brazilian & European) on the perception of Portuguese vowels and whether categorization patterns can predict difficulties in discrimination. Thirty-two beginner and elementary students of Portuguese as a Second Language (PSL) at California State University, Fresno, participated in the study. The results showed that vowel categorization and discrimination differed according to the dialect and language background. Heritage participants (H) had higher accuracy scores in their perception of Portuguese vowels, whereas bilingual participants exhibited similar discrimination patterns to monolinguals from previous research. Furthermore, perceptual overlap scores based on categorization percentages confirmed that categorization could predict difficulties in discrimination. This research sought to contribute to the understanding of heritage language acquisition, showing how language background can influence the perception of non-native sounds and the advantages of having contact with a heritage language, with potential implications for language teaching and learning strategies.

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THE PERCEPTION OF PORTUGUESE VOWELS BY BILINGUAL AND HERITAGE CALIFORNIAN ENGLISH LEARNERS OF PORTUGUESE

by

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TABLE OF CONTENTS

Page

LIST OF TABLES vii

LIST OF FIGURES viii

CHAPTER 1: INTRODUCTION 1

CHAPTER 2: LITERATURE REVIEW 3

Theoretical Models of Second Language (L2) Perception 3

Vowel Systems 7

Vowel Perception Studies 11

Bilingualism and Perception 15

Heritage Language and Perception 18

Research Questions 20

CHAPTER 3: METHODOLOGY OF THE PRESENT STUDY 22

Participants 22

Stimuli and Procedure 23

CHAPTER 4: RESULTS 27

Categorization Results 27

Perceptual Overlap 33

Discrimination Results 37

Perceptual Overlap and Discrimination 48

CHAPTER 5: DISCUSSION AND CONCLUSION 52

REFERENCES 61

APPENDICES 67

APPENDIX A: CONSENT FORM 68

APPENDIX B: BACKGROUND QUESTIONNAIRE 70

Page

APPENDIX C: SELF-DECLARED PROFICIENCY 76

LIST OF TABLES

Page

Table 1 *Categorization of BP (on the left) and EP Vowels (on the right) by Bilingual (ES) Participants in Percentages* 29

Table 2 *Summary of Categorization Patterns from Bilingual (ES) Participants, BP & EP Dialects.* 30

Table 3 *Categorization of BP (on the left) and EP Vowels (on the right) by Heritage Participants (H) in Percentages.* 31

Table 4 *Summary of Categorization Patterns from Heritage Participants (H), BP & EP Dialects.* 31

Table 5 *Predictions for Discrimination Difficulty Based on Perceptual Overlap Scores, for Each Group* 34

Table 6 *Predictions for Discrimination Difficulty Based on Perceptual Overlap Scores for Each Dialect* 36

Table 7 *Order of Difficulty for BP and EP for Bilingual (ES) and Heritage Speakers(H) and Percentage of Discrimination Accuracy* 39

Table 8 *Order of Discrimination Difficulty for Bilingual and Heritage Participants Based on Statistical Analyses.* 43

Table 9 *Result from the Fixed Effects Omnibus tests on Jamovi for Factors Influencing the Discrimination Accuracies* 45

Table 10 *Perceptual Overlap Scores and Discrimination Percentages Correct for Each Portuguese Vowel Contrast for the Bilingual (ES) Participants* 49

Table 11 *Perceptual Overlap Scores and Percentages Correct for Each Portuguese Vowel Contrast for the Heritage (H) Participants.* 50

Table 12 *Order of Discrimination of BP Vowels by Bilingual Participants as Elvin et al. (2018), and the Present Study* 54

Table 13 *Order of Discrimination of EP Vowels by Bilingual Participants as Elvin et al. (2018), and the Present Study* 54

Table 14 *Order of Discrimination of BP Vowels by Monolingual Participants and Bilingual Participants as in Vasiliev (2013), Elvin et al. (2018), and the Present Study.* 55

LIST OF FIGURES

Page

Figure 1 *Vowel Plot with Average Female and Male F1 and F2 Values for BP and EP.* 10

Figure 2 *A Sample Trial from the Categorization Task* 25

Figure 3 *A Sample Trial from the Discrimination Task* 26

Figure 4 *Categorization Percentage of Portuguese Vowels to English Categories* 32

Figure 5 *Discrimination Accuracy Scores on the Six BP and EP Contrasts for Bilinguals (ES) and Heritage Speakers (H)* 38

Figure 6 *Discrimination of BP and EP Contrasts by Bilingual (ES) and Heritage (H) Participants* 44

Figure 7 *Order of Discrimination Difficulty for BP and EP Vowels by Group* 45

Figure 8 *Order of Perceptual Overlap (PO) versus Discrimination (DI) Result – Bilingual (ES) Participants* 50

Figure 9 *Order of Perceptual Overlap versus Discrimination Result – Heritage (H) Participants* 51

CHAPTER 1: INTRODUCTION

Studies have shown that when learning a second language, individuals initially perceive and produce the target sounds differently due to the influence of their first language. Analysis of articulation and sound characteristics across different languages have been found to be predictive of challenges in perception, production, and recognizing words (see, for instance, Elvin, 2016; Elvin et al., 2018; Elvin et al., 2020; Vasiliev, 2013). This study aimed to examine how learners' auditory perception of non-native vowels affects discrimination, using the premises of the theoretical models of second language (L2) perception models (Best & Tyler, 2007; Elvin & Escudero, 2019; Flege & Bohn, 2021; Kuhl et al., 2008). Gaining insight into this process could guide the development of teaching strategies to address specific phonetic difficulties.

This work focused on the perception of vowels from two different Portuguese dialects, Brazilian (BP) and European Portuguese (EP), by two different groups, bilingual English/Spanish speakers (ES) and Heritage participants learners of Portuguese (H). The aim of the study was to investigate whether contact with a heritage language posed an advantage for the categorization and discrimination of vowels and whether vowels from different dialects affected them. Furthermore, it examined whether categorization could predict difficulties in vowel discrimination.

The experiment for the present study consisted of a vowel categorization task followed by a discrimination task, carried out in a computer lab at California State University, Fresno. Participants were university students taking Portuguese as a Second Language (PSL) classes at the beginner and elementary levels. Chapter 2 provides an overview of the theoretical frameworks that motivate the experiments in this study, models of second language speech perception, and a brief introduction to the phonological vowel systems of Brazilian Portuguese (BP), European Portuguese (EP),

and Californian English (CE). Chapter 3 describes the research methods for both experiments, and Chapter 4 reports their results. Lastly, Chapter 5 discusses the findings, proposes ideas for future studies, and concludes the thesis.

CHAPTER 2: LITERATURE REVIEW

Theoretical Models of Second Language (L2) Perception

Children learn their mother tongue more quickly and easily than adults learn a second language (Lightbown & Spada, 2021). Infants younger than nine months can typically discriminate speech sounds from all languages. At around age 9 to 12 months, infants become more sensitive to the phonetic contrasts in the language, ignoring "irrelevant" phonetic contrasts not used in their native language (Kuhl et al., 2008). Overall, language experiences from infancy through early childhood play a crucial role in shaping how individuals perceive speech sounds. This developmental trajectory is influenced by exposure to specific languages, social interactions, and educational opportunities, ultimately impacting language acquisition and literacy skills (Best & Tyler, 2007).

Upon reaching adulthood, their ability to discriminate non-native phonetic contrasts is not completely lost but significantly reduced. Achieving native-like proficiency in L2 skills such as grammar learning, lexical access, pronunciation, and production also becomes challenging. One of the main factors contributing to this difficulty is the influence of the native language, as at an initial stage of learning, their ability to perceive and produce new phonetic contrasts can be limited by their established L1 phonological system (see Elvin, 2016; Elvin et al., 2021; Tuninetti et al., 2020; Van Leussen & Escudero, 2015; Vasiliev, 2013). A number of theories, including the native language magnet model (NLM; Kuhl, 1993), the speech learning model (SLM; (Flege, 1995), the perceptual assimilation model (PAM; Best, 1995), and the second language linguistic perception model (L2LP; Escudero, 2005) have been proposed to explain how a person's native language can impact the acquisition of L2 sounds, word recognition, and production.

The NLM, and later the NLM-e model (Kuhl, 1993; Kuhl et al., 2008), uses the concept of "perceptual magnets" to explain why L2 learners hear a sound according to their L1 phonetic knowledge (Kuhl et al., 2008, p. 982). When infants learn their native language, they develop a "sound map" in their brains with language-specific representations, called prototypes, that help them discriminate a native sound clearly. Once these L1 prototypes are created in their memory, they work as "perceptual magnets," attracting similar members in that category. So, when an L2 sound is not part of the native category, learners map the nonexistent sound to one of their prototypes and create a new category. When compared with infants, adults can create new categories with more effort and time, as intrinsic factors, such as learners' L2 experience, age of L2 learning, amount of L2 use, quality and quantity of input, and type of instruction, and extrinsic factors, such as motivation, memory, and language learning aptitude, also interfere in the acquisition of a second language (Rato & Carlet, 2020).

The SLM and its revised version, SLM-r (Flege, 1995; Flege & Bohn, 2021), suggest that the phonetic categories in L1 shape how L2 sounds are processed. When learning an L2, learners classify L2 sounds into three categories in relation to L1: Same, New, or Similar sounds. Same sounds are easy to learn because they already exist in L1 and they are incorporated into an existing L1 category. New sounds are slightly harder to learn as they do not exist in L1, and a new category needs to be created. Lastly, similar sounds are the hardest to learn because they can be "classified" as more than one L1 sound, as listeners assume they are similar to both. The model emphasizes that the ability to create new phonetic categories decreases with age, as older learners are more likely to assimilate L2 sounds into existing L1 categories. In its revised version, SLM-r, the model suggests that the L1 phonetic similarity to an L2 sound dictates the facility to distinguish the L2 sound. Moreover, the revised model argues that, although there is a critical period

for language learning, phonetic learning continues throughout life, depending on the quality and the quantity of L2 input.

The PAM (Best, 1995) and its extension for L2 learners, PAM-L2 (Best & Tyler, 2007), similarly to SLM, account for how second language (L2) sounds are perceived in relation to native language (L1) categories and how the relationship between L2 sounds and the L1 phonological system influences the perception of L2 phonetic contrast. According to this model, listeners assimilate L2 sounds into native phonetic categories based on articulatory similarities or dissimilarities, which in turn affects their ability to accurately perceive and produce L2 sounds. In contrast to SLM, this model focuses on pairs of sounds (minimal pairs) and their differences and similarities with the L1 phonological system, which can determine the difficulty for learners in perceiving them. In this model, when learners hear a pair of sounds, they categorize it into an L1 category. PAM predicts discrimination difficulty in L2 based on articulation similarity with L1 and perception assimilation data. Different scenarios can arise depending on the similarities or differences of the sounds with the L1 phonological system.

The first scenario, called Two-Category (TC) Assimilation, occurs when the L2 contrast falls into two distinct existing L1 categories, and no problem with their perception is expected. In the Single Category (SC) Assimilation, both phones in the L2 contrast are similar to the same L1 category, leading to poor discrimination because the listener does not distinguish between the L2 sounds and categorizes both sounds under one phonological category from their L1. However, when both L2 sounds fit in the same L1 category, but one is perceived as a better exemplar, a Category-Goodness (CG) Assimilation scenario, discrimination is expected to be moderate as the easiest sound will be the one closer to the L1 category. Lastly, if neither of the L2 sounds fits into the listener's L1 inventory, an Uncategorized (U) scenario, it may be relatively difficult for them to learn the new sounds as they lack a reference point in the L1. With increased L2

experience, listeners may develop new categories or refine existing ones, leading to better discrimination over time.

The L2LP model (Elvin & Escudero, 2019; Escudero, 2005; Escudero et al., 2009) adds to this body of research by showing that a cross-linguistic acoustic analysis of L1 and L2 can predict more accurately difficulties in perception, word learning, and production at various developmental stages, from naïve to advanced proficiency levels (Elvin et al., 2021). Predictions are based on individual differences as they consider the stage in which each learner is (Elvin & Escudero, 2019). The L2LP model claims that comparing acoustic similarities can predict categorization patterns and discrimination difficulties, as discrimination difficulties depend on acoustic similarity and non-native categorization patterns (Elvin, 2016; Tuninetti et al., 2020). Elvin's (2016) study, for example, indicated that the L2LP model and the acoustic comparisons utilizing the participants' native speech data and the L2 production data is an effective method of predicting non-native discrimination and categorization patterns conducted both collectively and at an individual level. By comparing the acoustic characteristics of vowels in the native and target languages, predictions on how similarly vowels will be perceived (categorization) and how difficult they may be to distinguish (discrimination) can be made due to the relationship between acoustic properties and articulation of vowels (Elvin et al., 2014; Williams & Escudero, 2014; Elvin, 2016; Elvin et al., 2020, p. 72; Elvin et al., 2021).

The present study used the framework presented by PAM-L2 and L2LP, along with the idea that phonological and phonetic similarities can predict and explain discrimination difficulties of non-native contrasts, to identify the difficulties learners with different language backgrounds face when perceiving vowels from two different dialects BP and EP. Specifically, the study aimed to explore how Portuguese vowels from two different dialects were perceived by simultaneous bilingual Spanish/English speakers and

Heritage speakers of Portuguese, with English as their L1 and with contact with their HL but no fluency, from the Central Valley of California. This was done considering the phonological and acoustic variances between the L1, English, and L2, Spanish, to anticipate and comprehend perception challenges.

Vowel Systems

Vowels are the focus of perception studies, as they are produced with greater intensity and duration than consonants, providing more acoustic information than other segments. Additionally, vowels tend to be more variable than consonants across languages. This makes vowels an ideal area to determine whether and how learners' speech perception is influenced by language-specific mappings of the acoustic signal onto abstract phonetic categories between first and subsequently learned languages (Vasiliev, 2013).

Vowels can be classified based on three distinct features: vowel height (the height of the tongue and jaw when a vowel is produced, high, mid, or low), vowel backness (related to the position of the tongue, front, central, or back), and lip rounding (rounded or unrounded; Ladefoged & Johnson, 2015). Vowel sounds are distinguished by the pitches associated with their overtones, which form "three dark bands running horizontally through the vowel" (Ladefoged & Johnson, 2015, p. 23). The first formant (F1) values are related to the lower pitch, while the second formant (F2) values are related to the higher pitch. The frequency of these bands differs for each vowel, but it is the F2 value that distinguishes them.

Languages can differ phonologically, morphologically, and syntactically due to geographical, social, historical, and cultural factors. Chambers et al. (1998) defined dialects “as subdivisions of a particular language” and a language as “a collection of mutually intelligible dialects” (p. 3). Dialects can be classified at a broader level, within a

country, or at a narrower level, within a region or a particular social group, and their linguistic features make them part of a unique linguistic community. While Standard American English vowel inventory has 12 vowels, /i, ɪ, e, ɛ, æ, ə, ʌ, u, ʊ, o, ɔ, ɑ/ as in heed, hid, hate, head, had, but, hut, hoot, hood, hoe, haw, hard (C. Gorlston, personal communication, February 6, 2018), the vowel inventory from regional dialects differs in some aspects. Southern California English, for instance, has 10 steady-state vowels, /i, ɪ, e, ɛ, æ, ʌ, u, ʊ, o, ɑ/ in stressed position (Vasiliev, 2013). Compared to other dialects from the U.S., such as from the Northern and Middle Atlantic regions, Southern Californian vowels are generally more central (i.e., fronting the back vowels and backing the front vowels), which is the case for the CE vowel /u/, for example. There is also a tendency to unround the vowels, which is the case for the back vowels /u/ and /ʊ/ (Hagiwara,1997). The low-back vowel /ɑ/ in *caught* sounds like the mid-back vowel /ɔ/ in *cot*; the mid-front vowel /e/ has a lower first formant than /ɪ/, making it higher than the standard. The high back vowels are more front as they have a higher second formant, and because of that, the back vowels are more unrounded (Ladefoged, 1999; Ladefoged & Johnson, 2015). A recent change in Southern Californian English has lowered the vowel /o/ and a raised and backed /a/ (Grijalva et al., 2013).

Escudero et al. (2009) and Rato & Carlet (2020) showed that Portuguese from Brazil and Portugal share some similarities in their vowel inventory. Both dialects have a symmetric vowel inventory structure, with seven oral vowels in stressed position, /i, e, ɛ, a, ɔ, o, u/. Three vowels are unrounded front vowels (/i/, /e/, /ɛ/), the other three rounded back vowels (/o/, /u/, /ɔ/), and one central low vowel /a/. The front and back vowels form three pairings based on the similarity of their first formant (F1) value: two high vowels (/i/, /u/), two higher-mid vowels (/e/, /o/), and two lower-mid vowels (/ɛ/, /ɔ/). F1 was demonstrated to be affected by vowel height, gender, and dialect. For most speakers, Portuguese back vowels have higher F1 when compared to their corresponding front

vowels, so the F1 of /u/, for example, is higher than the F1 of /i/. F1 space is larger for Portuguese-speaking women than for men.

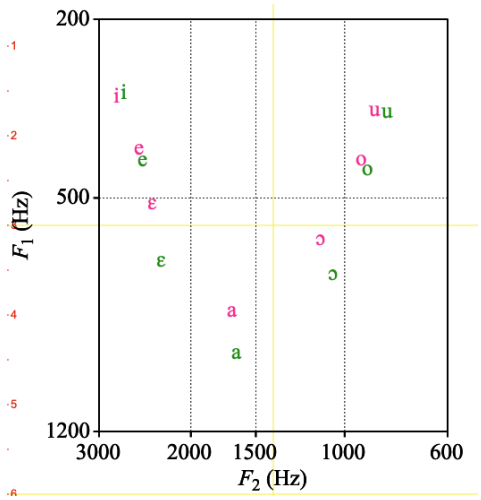
Portuguese vowels have no distinction in phonological length, but low vowels tend to have longer durations than high vowels (Escudero et al., 2009a). This effect is explained by the fact that open vowels need more jaw-lowering and, therefore, more time than closed vowels. Portuguese does not have a phonological tone either. Because of that, low vowels are expected to have a lower F0 than high vowels. There is a tendency, also presented in several Romance languages, to merge the lower-mid vowels with the higher-mid vowels in some regional varieties, and such a fact was also demonstrated regarding BP and EP. Like many languages, women tend to have longer vowel durations, larger vowel spaces, and higher F0, F1, and F2 values for females than males. Neither of the two dialects uses tone as a phonological feature, and no dialectal differences in F2 value were found (Chládková & Escudero, 2012).

However, EP has some features particular to its dialect, as Escudero et al. (2009a) described. First, EP oral vowels are shorter in duration than BP vowels, a fact that can make BP contrasts easier to discriminate. BP vowels /ɛ/, /a/ ad /ɔ/ have higher F1 values, and the lower-mid vowel /ɛ/ is higher and closer to /e/ in EP. Lastly, the vowel-intrinsic F0 effect is greater in BP than in EP. The acoustic plotting of BP and EP in Figure 1 uses the F1 and F2 formant values reported by Escudero et al. (2009, p. 1383). It can be observed that the BP and EP high vowels /i/, /e/, /u/, and /o/ are very close to each other due to their similar formant values, and because of that, they sound very similar in both dialects. On the other hand, the mid vowels /a/, /ɔ/, and /ɛ/ have more different formant values across the two dialects, being more distant from one another, and, acoustically, these EP vowels are higher than the BP ones. The contrast /e/-/ɛ/ is much closer in the EP variety than in the BP variety, suggesting that these two sounds might be merging into one sound. This contrast may sound more similar in EP and, therefore, more difficult to

perceive in EP than in BP. The contrast /o/-/u/ is very close in both dialects, making this contrast difficult to discriminate (Escudero et al., 2009; Elvin et al., 2018).

Figure 1

Vowel Plot with Average Female and Male F1 and F2 Values for BP and EP.



Note. Brazilian Portuguese *in green and European Portuguese in pink.* Adapted from "A cross-dialect acoustic description of vowels: Brazilian and European Portuguese," by P. Escudero, P. Boersma, A. S. Rauber, and R. A. Bion, 2009, *The Journal of the Acoustical Society of America, 126*(3), 1379–1393 (https://doi.org/10.1121/1.3180321).

Diaz-Granado's (2011) cross-linguistic acoustic comparison between American English and BP showed that BP /e/-/ɛ/ and English /e/-/ɛ/ are acoustically different, and the two BP sounds do not correspond to the English categories. Because of that, Americans were expected to create one single category when trying to produce both sounds. Remember that according to the PAM, when an L2 sound is similar to an L1 sound, the learner is more likely to acquire a native-like pronunciation of that sound. On the other hand, when a sound does not correspond acoustically to an L1 category, it will be difficult to produce. "As the American English and the BP low mid vowels belong to different phonetic entities, as far as the degree of openness is concerned, the speakers of

American English are not able to reproduce these Brazilian Portuguese vowels" (Díaz Granado, 2011, as cited in Smirnova Henriques et al., 2019, p. 68).

These vowels can be found in Portuguese in minimal pairs such as /e/-/ɛ/ pode (can) e pôde (could). Kendall (2004) explained that the mid-front vowel /e/ can cause pronunciation problems, as in English, this vowel is part of the diphthong /ei/, as in bait. Americans usually pronounce /ei/ when speaking Portuguese, where the /e/ sound should be isolated. Although the vowel /ɛ/ exists in English, such as in bet, the Portuguese version is more open than the English one, and Americans also tend to either pronounce the diphthong [ei] instead of [ɛ] or not open the vowel enough, making words such as fé (faith) commonly pronounced as /fei/.

The Portuguese vowel /o/ is longer than in English, and it has a different vowel quality vowel. In English, it also occurs in a diphthong in English, such as in boat, which makes Americans pronounce the isolated /o/ Portuguese sound as /ou/, such as the word boca (mouth) /ˈbo̯ukə/. These vowels can be found in Portuguese in minimal pairs such as /o-ɔ/ avô (grandfather) e avó (grandmother). Lastly, the phoneme /ɔ/ tends to be produced as /ɑ/, as Americans have difficulty producing the Portuguese mid-vowels.

Previous studies have demonstrated that acoustic variations from different dialects can impact perception, and experience with the language can affect the discrimination of vowels (Elvin et al., 2018; Elvin et al., 2021; Faris et al., 2016; Vasiliev, 2013). The present study aims to contribute to existing research by examining how bilinguals and heritage speakers perceive different dialects of Portuguese.

Vowel Perception Studies

Perception studies of L2 typically examine how learners perceive and distinguish non-native sounds to understand the influence of their native language on the perception of these sound words (see, for instance, Elvin, 2016; Elvin et al., 2018; Elvin et al., 2020;

Vasiliev, 2013). These studies often employ phoneme categorization and discrimination tasks. The categorization task assesses how listeners assimilate an L2 phoneme, e.g. a vowel, based on their L1 vowel inventory. At a group level, this task provides the percentage of listeners' classifications for each non-native vowel. The outcomes of the task involving the classification of non-native phonemes are expected to align with the results from cross-language phonetic acoustic similarity, as suggested by the L2LP model (Elvin & Escudero, 2019; Elvin et al., 2021; Escudero, 2005; 2009). In turn, the discrimination task evaluates how participants differentiate between contrasts by having participants identify which sounds are similar to the target sounds (Elvin et al., 2021; Faris et al., 2018; Vasiliev, 2013).

Some recent studies have investigated the perception of Portuguese vowels by different groups of participants. Kendall (2004) tested 20 beginning and advanced American university learners of Portuguese in perception and production tests, showing that both groups had problems identifying and categorizing the BP mid-vowels /e, ɛ, o, ɔ/. However, the advanced group presented better perception scores. Diaz-Granado (2011) also examined the assimilation and discrimination of BP mid-front vowel contrast /e/-/ɛ/ by twelve monolingual North American native speakers, seven monolingual English speakers learning Portuguese as L2, and five monolinguals with Spanish as their L2 and Portuguese as their L3. The result showed that the L2 and L3 learners of BP identified and discriminated these two mid-vowels better than the monolingual group, demonstrating that experience with the language helps the perception of different vowel categories. Moreover, L3 learners obtained higher discrimination accuracy scores than the L2 learners, showing that “the more varied the linguistic input the speakers have received, the better discriminative abilities they develop” (Díaz Granado, 2011, p. 205). Ultimately, both groups learned to differentiate the Portuguese mid-front contrasts, although they still had problems producing them.

Vasiliev (2013) assessed the initial state of assimilation and discrimination of all seven BP-stressed oral vowels by monolingual Californian English. In her discrimination task, the BP high-mid contrasts /a/-/ɔ/, /o/-/u/, and /e/-/i/ were the most difficult to discriminate for monolingual CE speakers. The /a/-/ɔ/ contrast was easier than /i/-/e/ but not significantly easier than /o/-/u/. The /i/-/e/ and /o/-/u/ contrasts were nearly equally difficult (Vasiliev, 2013, p. 89). The overlap scores calculated through the categorization scores were considered a strong predictor of discrimination difficulty, meaning that "the more listeners map a nonnative vowel contrast to overlapping native vowels, the more difficulty they have in discriminating this contrast" (Vasiliev, 2013, p. 102).

The BP contrasts /i/-/e/ and /o/-/u/ were also difficult to discriminate by functional monolingual Australian English and European Spanish speakers in Elvin et al. (2014). Acoustic cross-linguistic analysis showed that this difficulty could be explained by the fact that the two vowels presented in each BP contrast (L2) were acoustically similar to the same English (L1) category (for instance, BP /o/ and /u/ are both acoustically similar to Australian /ʊ/ and Iberian /u/, and BP /i/ and /e/ are both close to Australian /ɪ/ and Iberian /i/). These two BP contrasts are considered a SC assimilation resulting in an acoustic overlap and, therefore, difficulty in discrimination.

In a recent study by Elvin et al. (2018), the perception of the two Portuguese dialects, EP and BP, was investigated among naïve Californian English monolinguals and English-Spanish bilinguals. Both groups in the study categorized and discriminated the BP and EP vowels similarly, with no significant differences. In the discrimination task, there was also no significant difference in the discrimination of the BP and EP vowels, and the contrasts in both dialects were equally easy or difficult. For example, the contrasts /a/-/ɛ/ and /a/-/ɔ/ were the easiest to discriminate in both dialects, while the contrasts /i/-/e/ and /o/-/u/ were the most difficult. However, an interaction between vowel contrast and dialect was found, and some contrasts were easier to discriminate in

one dialect compared to the other; for instance, the BP contrasts /o/-/u/ and /i/-/e/ were easier to discriminate than the EP counterparts by both groups and the contrast /o/-/ɔ/ was easier in EP than in BP. Furthermore, the study demonstrated that the analysis of the acoustic differences between the two dialects correctly predicted the result of the categorization task and the difficulties in discrimination, and the perceptual overlap scores based on the categorization task showed a correlation with the discrimination difficulty.

When comparing the discrimination results by monolingual CE speakers for BP contrasts in Vasiliev's (2013) and Elvin et al.'s (2018) studies, the discrimination of BP vowels by monolingual speakers yielded very similar results. Like in Vasiliev, the /a/-/ɛ/ contrast in Elvin et al. (2018) was the easiest to discriminate, while /o/-/u/ and /i-e/ were the most difficult. There were some differences in the middle contrasts. In Vasiliev, the /e/-/ɛ/ contrast was the second easiest, with an accuracy score of 95%, followed by /o/-/ɔ/ with 94% and /a/-/ɔ/ with 82.5%. In Elvin et al. (2018), the /o/-/ɔ/ contrast was the second easiest to discriminate, followed by /e/-/ɛ/ and /a/-/ɔ/. The accuracy scores of Vasiliev were higher than those of Elvin et al. (2018).

Osborne (2021) is one of the few studies that has investigated the perception of the BP contrasts /ɛ/-/e/ and /ɔ/-/o/ by Heritage speakers of BP from New York, U.S. The choice for these two contrasts in the study was justified by the fact that these contrasts also pose some challenges for heritage speakers of similar Romance languages, such as Catalan and Galician. However, the author observed that these vowels are realized differently according to the dialect, and they are also the last contrasts to be produced by BP native speakers. In Osborne's study, the contrast /ɛ/-/e/ was more easily identified than the /ɔ/-/o/ contrast.

In sum, the studies on the perception of BP vowels by American speakers show that the high-mid vowels /e/, / ɛ/, /o/, and /ɔ/ are the most difficult to discriminate, and the

minimal pairs /e/-/ɛ/, /o/-/ɔ/, /e/-/i/, /o/-/u/ are the most challenging sounds for American English speakers. Acoustic analyses in previous studies (Elvin et al., 2018; Escudero, 2009; Vasiliev, 2013) confirm these contrasts as potential problems for categorizing and discriminating these vowels.

Bilingualism and Perception

Hoff (2014) defined bilinguals as people who are exposed to two languages in childhood. They may experience this dual exposure from birth (simultaneous bilinguals or bilingual first language learners) or acquire a second language later in childhood, (sequential bilinguals). When speaking a second native language, bilingual individuals posit a degree of interaction between their two native phonetic systems. Grijalva et al. (2013) showed that vowel production of Spanish and English vowels by bilingual Spanish/English speakers was more similar than those of the monolingual speakers, suggesting a degree of phonetic convergence. Diaz Granado (2011) demonstrated that the influence of bilingualism also affects perception. In her study, native English speakers learning BP as L2 and speakers who acquired Spanish as an L2 and then learned Portuguese as a third language (L3) outperformed the monolingual participants in the discrimination of BP mid-front vowels /e/-/ɛ/. Participants with Spanish as L2 and Portuguese as L3 performed better than monolinguals, showing that previous experience with a language that shares phonological similarities with Portuguese helped perception. The author, however, explains that the advantage of contact with additional languages is not straightforward:

> It is not the case that the more languages one learns, the easier it will be to produce a native-like sound in a new language, as it is commonly believed. It is claimed, however, that the longer the learner has been exposed to a specific sound, whether that sound was heard as part of one's first, second, or third

> language, the more likely it is that the learners will be able to reach native-like realization of that specific sound. (Diaz Granado, 2011, p. 218)

Kuhl et al. (2008) affirmed that bilinguals might perceive better because they can access two vowel systems to help them categorize and discriminate new sounds. Due to the requirement of switching between languages, bilingual children also show better attentional processes. Although bilingual children sometimes learn new words more slowly than monolingual children, as they need to access two "databases," this disadvantage disappears later, and bilingual children perform in both languages just like monolinguals.

Nonetheless, the idea that bilingual speakers may have an advantage in perceiving foreign vowels better when compared to monolinguals is controversial, and bilinguals are believed to behave similarly to monolingual speakers. For example, the study developed by Grijalva et al. (2013) compared the vowel production of five American English vowels by bilingual Spanish-English speakers and monolingual Southern CE speakers, showing that bilingual and monolinguals differed in their production, and bilingual participants produced tokens close to the phonetic categories of each of their languages but not quite the same with either. This study validated the general belief that people fluent in two languages tend to produce sounds similar to the distinct phonetic categories of each language but not identical to either.

Smirnova Henriques et al. (2019) investigated image identification, vowel identification, and word discrimination of the BP contrasts /ɛ/-/e/ and /ɔ/-/o/ by native speakers of Russian living or have lived in Brazil with different Portuguese proficiency, finding that late Russian-Portuguese speakers did not exhibit better discrimination of Portuguese mid-vowel contrasts /ɛ/-/e/ and /ɔ/-/o/ compared to monolingual native Russian speakers. In this case, factors such as language instruction and length of residence did not impact their ability to perceive these sounds accurately, and the ability

to create new phonetic categories is not directly dependent on the length of contact with the target language but rather on the quality and quantity of input.

Similar discrimination result of BP vowels between monolingual and bilingual participants was also reported by Elvin et al. (2018). The discrimination task results indicated that individuals' language background (monolingual Californians and bilingual English-Spanish speakers) did not significantly impact their perception of BP and EP vowels, as expected. The only significant finding related to the interaction between language background and vowel contrast was in relation to the /o/-/u/ contrast, which was more accurately discriminated by bilinguals than monolinguals. This difference could be attributed to the fact that this contrast is more similar to Spanish across both dialects. However, this advantage for bilinguals was not significant.

While contact with other languages does not seem to help the perception of novel language contrasts, the amount of exposure to the target language can positively affect it. For example, Rato & Carlet (2020) investigated how EP learners of English categorized six British English vowels and which learner variables could predict perceptual performance, such as the age of onset of learning, length of formal instruction, amount of L2 use, and vocabulary size. The result demonstrated that the most significant factor that positively affected vowel identification was the amount of daily L2 use. Smirnova Henriques et al. (2019) also showed that the quality and quantity of input positively affected the discrimination of the mid-BP contrasts by Russian naïve speakers living in Brazil. Overall, “the capacity of creating a new phonetic category does not depend directly on the time of residence, but on the quality and quantity of input” (Smirnova Henriques et al., 2019, p. 81).

Heritage Language and Perception

HS are learners who acquire or learn their heritage language (HL) during childhood in a migrant environment. Usually, they have one or two parents or grandparents who are native speakers of their HL, "the language of the culture in which they were born" (Hoff, 2014, p. 262). The contact with the HL can start at birth or in the early years (up to age 3 or 4). In this case, heritage speakers learn their HL in a naturalistic setting, speaking their HL daily at home. For this, they were considered simultaneous or early successive bilinguals. Their HL is their L1 or one of the two L1 (usually the majority language from the current country). Sometimes, their HL becomes the weaker language later in life (after age 4/5), and their societal language (the majority language) becomes the dominant language, making them unbalanced bilinguals. For a long time, research has included heritage speakers in the group of bilingual speakers. However, recent studies have shifted this approach. HS is considered "a particular type of bilingual speaker" (Flores, 2015, p. 251), usually second-generation immigrants, with the host country language as their dominant language and differing HL development and proficiency. Therefore, heritage speakers are studied as a particular speaker profile rather than treated as bilingual. Flores (2015) emphasized that the similarities between bilingualism and heritage language cannot be disregarded, and studies on bilingual language acquisition should not be ignored in heritage language research.

The study of the phonetic ability of heritage speakers is an area that has not been widely explored. A few studies on heritage speakers' perception of vowels show that heritage speakers discriminate vowels similarly to native speakers of the HL. Osborne (2021), for example, found that H of BP born and/or raised in New York, US, with low English dominance (LED) discriminated the BP mid-vowel contrasts /e/-/ɛ/ and /o/-/ɔ/ similarly to native Portuguese speakers, indicating that both groups are equally proficient in this aspect of the language. However, Higher English dominant (HED) heritage

speakers had more difficulty identifying vowels in the contrast /ɔ/-/o/ than the /ɛ/-/e/ contrast compared to the LED group. This result suggests that the amount of English dominance and the use of Portuguese may interfere with the identification of vowels by H. Those who spoke more Portuguese and did not have a high proficiency in English perceived the vowels similarly to native Brazilian speakers, suggesting that language dominance in English may reduce the use of Portuguese, which could also result in a lower perception of some contrasts. However, H are expected to have better linguistic competence than L2 adult learners, as they have the advantage of having early contact and a higher amount of exposure to the HL (Flores & Rato, 2016).

The benefits of early exposure to a HL have been demonstrated in studies involving international adoptees as well. For instance, Choi et al. (2017) found that South Korean children adopted at a very young age (between 6 months and 1 year) by Dutch-speaking families in the Netherlands retained implicit phonological knowledge of their HL, despite losing all active contact with it. Similarly, Zhou et al. (2021) examined Mandarin Chinese adoptees raised by comparable families and observed that they were better at distinguishing tonal differences in Mandarin than reproducing them. Both studies support the concept of a critical period for language acquisition, during which early exposure leaves enduring neural imprints. These imprints can facilitate faster recognition and comprehension of the HL, even years after disuse.

The present study aims to contribute to the research on second language acquisition by investigating the non-native perception of Portuguese vowels among bilingual English and Spanish speakers, as well as Heritage English speakers from the Central Valley in California. The heritage participants in this study did not speak the HL fluently, but they have been exposed to it since childhood within their home environment. Considering the definition of heritage speakers discussed earlier, the participants could not be considered heritage speakers but rather heritage learners of the HL. 2 of the 10

heritage participants identified as monolinguals, while two stated that Portuguese is their second L1 language. However, they reported being more fluent in English, ranking their proficiency in Portuguese as medium to low (3 or 2 out of 5).

Research Questions

The present research aimed to examine how English-speaking heritage and ES bilingual learners of EP categorize and discriminate BP and EP vowels in an attempt to answer the following research questions (RQ):

RQ1: Which Portuguese contrasts are more difficult for bilinguals and heritage learners to categorize?

Based on previous research, the contrast /a/-/ɛ/ is expected to be the easiest to discriminate, and the contrasts /o/-/u/ and /i/-/e/ to be the hardest (Vasiliev, 2013; Elvin et al., 2018; Osborne, 2021).

RQ2: Does the perception of BP and EP differ?

Given that previous findings indicate dialect matters regarding the perception of L2 languages (Elvin et al., 2018), it is predicted that bilinguals and heritage learners will perceive BP and EP contracts differently. Although the Portuguese mid-vowels /o/-/u/ and /i/-/e/ are expected to be more difficult to discriminate in both dialects, the order of discrimination difficulty may vary, considering that the dialects have acoustic differences that interfere with the categorization and discrimination.

RQ3: Do ES bilinguals and heritage learners of EP differ in their perception of BP and EP vowels?

Previous studies have shown that bilingual ES speakers perceive vowels similarly to monolingual English speakers (Elvin et al., 2018), while heritage participants show similar perceptual abilities as native speakers of the HL. This suggests that familiarity and language exposure to the HL, even without being proficient, are advantageous for the

perception of the HL sounds (Flores & Rato, 2016; Osborne, 2021). Therefore, heritage learners of EP are expected to perform better than ES bilinguals in discriminating both dialects. Moreover, considering that both groups have been exposed to EP dialect more than BP, both groups are expected to discriminate EP vowels better than BP.

CHAPTER 3: METHODOLOGY OF THE PRESENT STUDY

Two different experiments were conducted to investigate how ES bilingual and Heritage learners of EP perceive BP and EP vowels: categorization and discrimination. This chapter describes the participants, the stimuli used in the experiments, and the experimental procedures.

Participants

32 Californian English speakers were recruited from the beginner and elementary EP classes offered at California State University, Fresno, and from the Portuguese community around the area (22 females and 10 males, mean = 20 years, age range: 18-39 years). They were bilingual ES speakers with basic Portuguese knowledge and heritage learners of EP. Students were granted extra credits for participating in the experiment, and the community members volunteered their time without compensation. All participants provided informed consent, and an online background questionnaire was administered using Qualtrics prior to completing the experiments (see Appendices A and B).

Ten participants, five females and five males, were second or third-generation heritage language learners of EP. Participants reported that they do not speak Portuguese fluently but have heard it in their homes. The participants ranged in age from 18 to 39 ($\tilde{X}$=22). They were born in the Central Valley of California, U.S., and had English as their L1. 7 H participants were undergraduate students at California State University, Fresno, taking Portuguese classes, while the other three were members of the Portuguese community around the area. All but one declared themselves monolinguals, but two participants (n. 2 and 3) declared Portuguese as their native language along with English, being more fluent in English but having a high knowledge of BP. They all identified themselves closely to the Azorean and/or Brazilian Portuguese community. The quality

and quantity of the exposure of HL varied. Nine had at least one parent speaking EP or BP, while one had just grandparents speaking the HL. Their self-declared proficiency also varied widely (see Appendix C). One H participant declared a beginner-level proficiency in all five skills, while five H participants self-declared a low level of Portuguese dominance among the five skills (1-2 being low, 3 being neutral, and 4-5 being high, they self-ranked their skills as three or below). Three ranked at least one of the five skills as having a high proficiency level (4 out of 5). Two participants did not declare their proficiencies.

Twenty-two participants, 18 females and four males, were self-declared as bilingual ES speakers from birth (two participants identified as monolingual but stated Spanish as their L1). All but one ranked their Spanish proficiency as 3 or above (on a scale from 1 to 5), with at least one of the five skills rated as 5 (from 1 to 5). They ranged in age from 18 to 22 ($\tilde{X}$=19). They were born primarily in the Central Valley of California, U.S., and other areas of California. They were all undergraduate students enrolled in Portuguese classes at Fresno State. Just nine bilingual participants self-declared their Portuguese proficiency. They reported having some knowledge of the language, maybe because some had already taken Portuguese classes in high school and university. The similarity between the Spanish and Portuguese languages may also give them the confidence to report a higher Portuguese proficiency. See Appendix C for more details about the participants' self-declared proficiency.

Stimuli and Procedure

Stimuli

To reliably check if the perception was influenced by dialect, participants perceived both European and Brazilian Portuguese vowels produced by 10 male and 10 female speakers of BP from São Paulo and 10 male and 10 female speakers of EP from

Lisbon, Portugal. The EP and BP vowel tokens were the same in the categorization and the discrimination tasks and were selected from Escudero et al. (2009) and used by Vasiliev (2013) and Elvin et al. (2018). The software Praat (Boersma & Weenink, 2018) was used to develop the experiments. All participants were tested in a computer lab at California State University, Fresno, using desktops from the University and headphones provided by the researcher.

The tokens extracted from pseudo-words with two syllables in the /fVfe/ context, where "V" was the stressed syllable with the target vowel corresponding to one of the seven Portuguese vowels /i, e, ɛ, a, o, ɔ, u/ (for a discussion about the two-syllable context, see Strange et al., 2007). The tokens were extracted from a native production task in which a native speaker read a carrier sentence with the pseudo-words e.g., "Fêfe. Em fêfe e fêfo temos ê" which translates to: "Fêfe. In fêfe and fêfo we have ê" (Escudero et al., 2009)

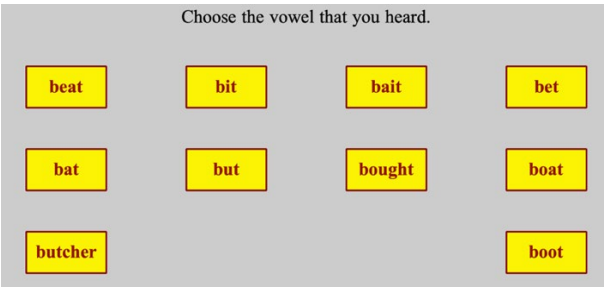
The voiceless fricative context /fVfe/ was chosen because other contexts, such as voiceless stops, differ in the VOT (voice onset time) and formant transitions (Vasiliev, 2013). The choice to present vowels in isolation as tokens was made because when vowels are presented alone, listeners must rely on their abstract representations of the target vowels. This helps prevent contextual factors that may affect vowel categorization and discrimination (Elvin et al., 2018). Harrington (2010) explained that vowels are dynamic and cannot be distinguished only by the information presented in the vowel target. Vowels extracted from the CVC context instead of isolated V syllables help identify more accurately (Harrington, 2010). Moreover, "silent center syllables" vowels (those extracted from the CVC context) were shown to be as well identified as when presented in the original CVC syllables.

Categorization Task

In the categorization task, 70 BP and 70 EP tokens were used as stimuli in a random order. Participants were asked to listen to an item and label the sound to one of the ten CE vowels. The options on the screen, “beat, bit, bait, bet, bat, but, bought, boat, butcher and boot” corresponded to the ten CE vowel phonemes /i, ɪ, ɛ, ɜ, a, ɑ, ʌ, ɒ, ɔ, ʊ, u/ in the /bVt/ context. Each listener was asked to listen to the vowel sound and click on the word with a vowel most similar to the vowel they had heard. Participants were instructed to choose a response even if unsure. They first completed a short practice session to become familiar with the task and the stimuli. The between-trial interval was one second. Participants were instructed that they could take a break at any moment. Participants took around of 30 minutes to complete the test. Figure 2 shows a sample screenshot of the categorization task.

Figure 2

A Sample Trial from the Categorization Task



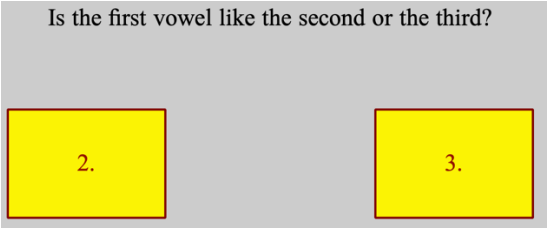
Discrimination Task

The discrimination task was a two-alternative forced choice (2AFC) task in the XAB format following previous studies (Diaz-Granado, 2011; Elvin et al., 2018; Escudero & Wanrooij, 2010; Escudero & Williams, 2012; Vasiliev, 2013). It targeted the discrimination of six BP and EP vowel contrasts, namely /a/-/ɔ/, /a/-/ɛ/, /i/-/e/, /o/-/u/, /e/-/ɛ/, and /o/-/ɔ/. There were 504 different auditory sounds in this task. The sound with the

target vowel (X) was presented in each trial with two other sounds (A and B). A and B stimuli were always from different vowel categories from X and counterbalanced, appearing in XAB or XBA in a randomized order. Participants were required to decide whether the first word (X) they heard sounded more like the second word (A) or the third (B) by choosing on the screen the number "2" for the second sound or "3" for the third sound. They were informed that they should respond as fast as they could as the response times would be collected. As in the categorization task, participants familiarized themselves with the task during the training phase. The task took around 40 minutes, and participants could take a break anytime. Figure 3 shows a sample screenshot of the discrimination task.

Figure 3

A Sample Trial from the Discrimination Task



Chapter 4 will present the results of the categorization and discrimination tasks of BP and EP vowels by ES bilingual and heritage learners of EP.

CHAPTER 4: RESULTS

This chapter presents the results of the two experiments, categorization and discrimination, showing how 32 participants, divided into two groups, ES bilingual and heritage learners of EP, differed in their perceptions of the BP and EP dialects.

Categorization Results

Remember that the SLM and SLM-r (Flege & Bohn, 2021), the PAM -L2 (Best & Tyler, 2007), and the L2LP (Elvin & Escudero, 2019) model claim that the difficulty level in discriminating a target sound can be explained based on how the L2 sound is assimilated into the native categories. If an L2 vowel sound is similar to an existing L1 vowel, this phone will be easy to discriminate because it already exists in their L1 phonetic inventory. However, when the L2 sound does not exist in the L1 inventory, new categories will need to be formed, so initially, this sound will be difficult for learners. Lastly, if the L2 sound is similar to two or more L1 sounds, it will be the hardest to discriminate, as it can be classified as more than one L1 sound.

Predictions can be made regarding discrimination difficulties depending on the scenario created by the L2 contrasts. Very good discrimination is expected when the two L2 sounds in a contrast are categorized as two different L1 sounds, forming a TC assimilation in the PAM and a Similar Learning Scenario in the L2LP model. Good discrimination is also expected when Uncategorized-Categorized assimilation in the PAM occurs, that is, when one of the L2 sounds is perceived as an L1 category and the other does not exist in the L1 inventory. On the other hand, discrimination difficulty is predicted to happen in two different scenarios. The first scenario occurs when both sounds in the L2 contrast are mapped into the same L1 category, creating a Single-category assimilation in the PAM or a New Scenario in the L2LP model. The other difficult scenario happens when one of the two L2 sounds in a contrast can be categorized

into two or more L1 categories, creating a Multiple-category assimilation in the PAM or a subset scenario in the L2LP (Elvin & Escudero, 2019).

In the categorization task, participants listened to BP and EP vowels one at a time and classified them as one of the ten CE vowel sounds on the screen. The result from the categorization in the present study is expected to show that discrimination difficulty can be predicted depending on how the Portuguese vowels are categorized into L1 vowels. ES and H participants are expected to show some differences in their categorization patterns, as ES participants were shown to perceive vowels similarly to monolinguals, and language experience and exposure to the HL was shown to help with perception (Elvin, 2024; Elvin et al., 2018; Flores & Rato, 2016; Rato & Carlet, 2020). Lastly, some differences between the categorization of the BP and EP dialects are also expected due to acoustic and phonetic differences between the two dialects explained by the models of L2 speech perception cited above (Elvin et al., 2018; Escudero, 2009).

Examining the categorization from the ES bilingual participants in Table 1, each BP and EP vowel was similar to more than one CE vowel. There were differences in the categorization choice depending on the dialect. BP and EP /i/ were categorized into CE /i/ and /ɪ/, but BP /i/ was mapped to CE /i/ 45% of the time, and as CE /ɪ/ 38%, whereas the categorization pattern was reversed for EP /i/, being categorized as CE /i/ 37% of the time and as CE /ɪ/ 43%. BP /e/ was categorized into four different CE vowels, namely /i/, /ɪ/, /e/, and /ɛ/, but the same EP vowel was not categorized as CE /e/. BP /ɛ/ was categorized as CE /e/, /ɛ/, and /æ/, but the same EP vowel was categorized as two CE categories, /ɪ/ and /ɛ/. BP and EP /a/ were categorized into CE /æ/, /ʌ/, and /a/, with higher percentages for CE /æ/ in BP (51%) and for CE in EP (43%). The vowel /o/ was mainly mapped into CE /o/ in the two dialects (43% in BP and 33% in EP) and also into CE /ɑ/ and /ʌ/. The EP /o/ was also categorized as CE /u/, which didn't happen with the same BP vowel. BP /ɔ/ was mostly mapped to CE /ɑ/ 45% of the time, followed by /ʌ/ and /o/. In EP, the

same vowel was mostly mapped as CE /o/ 40% of the time, followed by /ɑ/ and /ʌ/. Lastly, BP and EP /u/ were similarly categorized mostly as CE /u/, 47% of the time in BP and 61% in EP.

Table 1

Categorization of BP (on the left) and EP Vowels (on the right) by Bilingual (ES) Participants in Percentages

ES	Brazilian and European Portuguese						
	i	e	ɛ	a	o	ɔ	u
i	**45** / 37	10 / 11					
ɪ	38 / **43**	**37** / **41**	-/14				
e		15/-	10/-				
ɛ		26 / 22	**57** / **44**				
æ			13/-	**51** / 33			
ʌ				26 / **43**	12 / 11	26 / 10	
a				15 / 14	31 / 25	**45** / 39	18 / 11
o					**43** / **33**	15 / **40**	20 / 11
ʊ							
u					-/18		**47** / **61**

Note. The Native Vowel Category with the Highest Classification Percentage Appears in bold.

The summary of the categorization patterns for both dialects performed by the ES participants is shown in Table 2. Some categorization overlaps occurred between the BP and EP vowels, as two vowels were categorized as the same L1 vowels. For example, BP /i/ and /e/ were both categorized as CE /i/ and /ɪ/, and BP /a/ and /o/ were categorized as CE /ʌ/ and /ɑ/, although in different proportions. The most significant categorization overlaps were observed between the EP vowel pairs /i/ and /e/ and EP /o/ and /ɔ/. /i/ and /e/ were both mainly categorized as CE vowel /ɪ/ (/i/ 43% of the time and /e/ 41% of the time), and /o/ and /ɔ/ had the highest categorization percentages as CE /o/ (/o/ 33% and /ɔ/ 40%). This indicates that these EP vowel pairs sounded very similar to the same CE vowel.

Table 2

Summary of Categorization Patterns from Bilingual (ES) Participants, BP & EP Dialects.

BP	ES
i	i ɪ
e	ɪ ɛ e i
ɛ	ɛ æ e
a	æ ʌ ɑ
o	o ɑ ʌ
ɔ	ɑ ʌ o
u	u o ɑ

EP	ES
i	**ɪ** i
e	**ɪ** ɛ e i
ɛ	ɛ ɪ
a	ʌ æ ɑ
o	**o** ɑ u ʌ
ɔ	**o** ɑ ʌ
u	u o ɑ

Note. The most significant overlaps are shown in bold.

When analyzing the categorization by the Heritage (H) participants, each BP and EP vowel was also categorized into more than one CE vowel, with the classification percentage indicating a more significant categorization of one vowel. BP /i/ was mapped 76% of the time as CE /i/, and the same EP vowel was also mainly classified as CE /i/, albeit less consistently (42% of the time). BP and EP /e/ were predominantly categorized as CE /ɪ/ (33% and 39% respectively), but also as CE /ɛ/ (25% & 17%), as /e/ (12% & 13%), and as /i/ (21% & 22%). BP and EP vowel /ɛ/ were classified mostly as CE /ɛ/ (63% & 54%); however, the BP vowel was also categorized as /æ/ (10%) and /e/ (11%), while the EP vowel was categorized equally as /ɪ/ and /e/ (14%). BP and EP /a/ were most categorized as CE /æ/ (49% and 37% of the time), followed by CE /ɑ/ (25% & 27%) and /ʌ/ (14% & 23%). The BP vowel /o/ was mapped to three different CE categories: /o/ (51%), /u/ (17%), and to a lesser extent, /ɑ/ (12%), while the same EP vowel was categorized as CE /u/ (36% of the time), /o/ (33%), and /ʊ/ (13%). BP and EP /ɔ/ were mostly mapped to CE /ɑ/ (53% & 41%); however, in BP, it was also mapped as CE /ʌ/ (16% of the time) and /o/ (10%), while EP was mapped to CE /o/ (30%). Lastly, BP and EP /u/ were most categorized as CE /u/ (57% of the time in BP and 54% in EP); however, in BP, the same vowel was also classified as CE /ʊ/ (21% of the time), while in EP, it was also classified as CE /ʊ/ (23%) and as /o/ (10%). The categorization patterns for the BP

and EP vowels are presented in Table 3, and the summary of the categorization patterns is shown in Table 4.

Similarly to the categorization patterns of the ES participants, the H participants' categorization patterns overlapped between some pairs of BP and EP vowels. However, significant categorization overlap was observed only between the EP pairs /o/ and /u/, as they were both mainly classified as CE /u/ (EP /o/ 36% of the time and EP /u/ 54%).

Table 3

Categorization of BP (on the left) and EP Vowels (on the right) by Heritage Participants (H) in Percentages.

H	Brazilian and European Portuguese						
	i	e	ɛ	a	o	ɔ	u
i	**76/42**	21/22					
ɪ	11/33	**33/39**	-/14				
e		12/13	11/14				
ɛ		25/17	**63/54**				
æ			10/-	**49/37**			
ʌ				14/23		16/-	
ɑ				25/27	12/-	**53/41**	
o					**51**/33	10/30	-/10
ʊ					-/13		21/23
u					17/**36**		**57/54**

Note. The Native Vowel Category with the Highest Classification Percentage Appears in bold.

Table 4

Summary of Categorization Patterns from Heritage Participants (H), BP & EP Dialects.

BP	H
i	i ɪ
e	ɪ ɛ i e
ɛ	ɛ e æ
a	æ ɑ ʌ
o	o u ɑ
ɔ	ɑ ʌ o
u	u ʊ

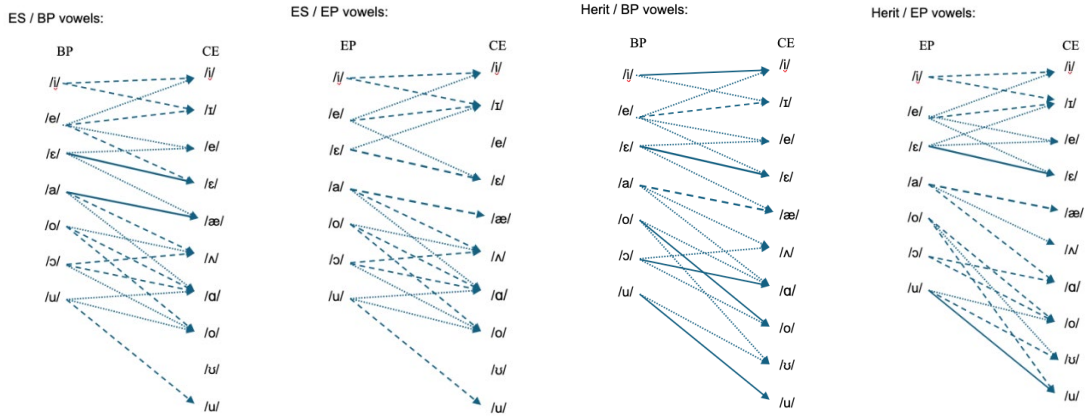
EP	H
i	i ɪ
e	ɪ i ɛ e
ɛ	ɛ e ɪ
a	æ ɑ ʌ
o	**u** o ʊ
ɔ	ɑ o
u	**u** ʊ o

Note. The most significant overlaps are shown in bold.

Some differences were observed between the categorization patterns among the participants and the dialects. Some BP and EP vowels had the highest classification percentages among different groups. Overall, H participants had higher categorization percentages. For instance, the higher percentages for BP /i/ were 76% by H and 45% by ES; BP /ɛ/ also had higher percentages by H participants than by ES (63% vs. 57%). The only BP vowel with a higher categorization score by ES was BP /e/, with a score of 37% vs. 33%, meaning ES participants were more certain when classifying this vowel. When comparing the categorization results among the two dialects, the EP vowels showed less consistency among the two groups of participants, with most vowels presenting a higher percentage in BP than in EP. This is the case, for instance, between the highest classification percentages for BP and EP /i/ among ES participants, with 45% in BP and 43% in EP; between BP and EP /ɛ/, with 57% and 44%; or between BP and EP /a/, with 51% and 43%. Among H participants, the vowel /i/ had 76% in BP and 42% in EP; /ɛ/ had 63% in BP and 54% in EP; and /a/ had 49% in BP and 37% in EP. Figure 4 summarizes the categorization patterns of BP and EP by ES and H participants.

Figure 4

Categorization Percentage of Portuguese Vowels to English Categories



Note. Solid Arrows Above 51%, Dashed Arrows 26-50%, Dotted Arrows 10-25%.

To check if dialect and language background statistically influenced the classification of the Portuguese vowels, a chi-square (X^2) test of independence was conducted on Jamovi software (2024) to examine the association between language participant group and dialect. The results all showed p-values less than 0.001, indicating that vowel type significantly influenced participants' responses and that these responses were not due to random chance. When examining the effect of language background and dialect in the categorization of the Portuguese vowels, their extremely small p-values (all < 0.001) indicated a strong and highly significant association between language background and dialect, showing that the vowels were categorized differently by the ES and H participants, and the categorization varied depending on the dialect.

With the categorization results now established, the subsequent section explores the predictions concerning their accuracy within a categorical XAB discrimination task utilizing identical vowel stimuli. Furthermore, it examines the implications of phonetic assimilation patterns as indicators of challenges in discrimination.

Perceptual Overlap

The PAM and the L2LP model propose that discrimination difficulty happens because of perceptual similarity. Perceptual overlap occurs when two non-native sounds of a non-native contrast are perceived as the same native language vowel or set of vowels, leading to difficult discrimination. On the other hand, if there is no overlap between a pair of vowels, each non-native sound is identified as having different native language categories, resulting in excellent discrimination accuracy (Elvin et al., 2018; Elvin et al., 2021; Faris et al., 2018; Vasiliev, 2013; Wang et al., 2022).

Following the “cross language assimilation overlap” method adopted by Levy (2009), the percentage of perceptual overlap scores for the six BP and EP contrasts were calculated to check if perceptual overlap scores can predict discrimination accuracy. A

quantitative overlap score is obtained by adding the smaller percentages of each overlapping category. For example, for the categorization scores of BP contrast /a/–/ɔ/ shown in Table 5, H participants classified the BP /a/ as CE /æ/ 49%, /ɑ/ 25%, and /ʌ/ 14% of the time. BP /ɔ/ was also classified as CE /ɑ/ 53%, /ʌ/ 16%, and /o/ 10% of the time. The overlapping vowels in this contrast were CE /ɑ/ 25%, and /ʌ/ 14% of the time for BP /a/ and CE /ɑ/ 53%, /ʌ/ 16% for BP /ɔ/. Adding the smaller overlapping percentages of both vowels, BP /a/ 14 % and BP /ɔ/ 25%, the perceptual overlap score would be 14+25 = 39%. Following previous studies (Elvin et al., 2018; Vasiliev, 2013), perceptual overlap scores below 10% were excluded from the calculation.

Table 5

Predictions for Discrimination Difficulty Based on Perceptual Overlap Scores, for Each Group

ES	Easiest					Most difficult
BP	/a/–/ɛ/ < /e/–/ɛ/ < /o/–/u/ < /a/–/ɔ/ < /i/–/e/ < /o/-/ɔ/					
Overlap score	13	36	38	41	47	58
EP	/a/–/ɛ/ < /a/–/ɔ/ < /e/–/ɛ/ < /o/–/u/ < /i/–/e/ < /o/-/ɔ/					
Overlap score	0	25	36	40	52	69
H	Easiest					Most difficult
BP	/a/–/ɛ/ < /o/–/u/ < /o/-/ɔ/ < /i/–/e/ < /e/–/ɛ/ < /a/–/ɔ/					
Overlap score	10	17	22	32	36	39
EP	/a/–/ɛ/ < /a/–/ɔ/ < /o/-/ɔ/ < /e/–/ɛ/ < /i/–/e/ < /o/–/u/					
Overlap score	0	27	30	44	55	59

Note. '<' means 'easier than'.

Table 5 also shows the predictions for the discrimination difficulty based on the overlap score for each BP and EP contrast organized from the lowest to the highest score. A score of 0% indicates no overlap between the contrasts; the closer the score is to zero, the easier the contrast should be discriminated, as the two vowel sounds are categorized into two different L1 categories. On the other hand, the closer the score is to 100%, the

more difficult it should be to discriminate, as listeners might perceive both sounds in the contrast as belonging to the same single or multiple categories.

For the ES participants, the contrasts /o/-/ɔ/ and /i/-/e/ had the same two highest perceptual overlap scores for both dialects, with 58% in BP and 69% in EP, and 45% in BP and 51% in EP, respectively. The third hardest contrast varied in each dialect; the contrast /a/-/ɔ/ had 41% in BP, followed by /o/–/u/ with 38%, /e/–/ɛ/ with 36%, and /a/–/ɛ/ with 13%. The /o/-/u/ contrast has 40% of overlap in EP, followed by /e/–/ɛ/ with 36%, /a/–/ɔ/ with 25%, and /a/–/ɛ/ with 0%. In both dialects, the contrast /a/-/ɛ/ is expected to be the easiest for both groups of participants to discriminate.

Turning to the H participants, the contrast /a/-/ɛ/ was also predicted to cause no problem for discrimination in both dialects, with a 10% overlap score for BP and 0% for EP. In BP, the contrast /a/-/ɔ/ received the highest overlap score of 39% in BP, followed by /e/-/ɛ/, with 36%, /i/-/e/ with 32%, /o/-/ɔ/ with 22%, and /o/-/u/ with 17%. In EP, the contrast /o/-/u/ had the highest overlap score of 59%, followed by /i/-/e/ with 55%, /e/-/ɛ/ with 44%, /o/-/ɔ/ with 30%, and /a/-/ɔ/ with 27%.

Some contrasts had a higher perceptual overlap score in one dialect than the other. Following the suggestion from Elvin et al.'s (2018) study, a 5% difference was considered a threshold for determining a difference in perceptual overlap between dialects. For example, among the H participants, the contrasts /a/-/ɛ/ and /a/-/ɔ/ were higher in BP, whereas the contrasts /i/-/e/, /o/-/u/, /e/-/ɛ/, and /o/-/ɔ/ had a higher percentage in EP. Among the ES participants, the contrasts /a/-/ɛ/ and /a/-/ɔ/ were higher in BP, whereas the contrasts /i/-/e/ and /o/-/ɔ/ had a higher percentage in EP. The contrast /e/-/ɛ/ did not have a 5% difference among both dialects, with an overlap score of 38% in BP and 40% in EP, meaning that the discrimination would have similar results in both dialects.

Therefore, according to the perceptual overlap scores, the discrimination between the BP and EP contrasts was expected to differ among the participant groups. Although all participants were likely to agree on the easiest contrast, the ES group was expected to have the most consistent discrimination scores. This is because the order of discrimination varies only among the middle contrasts, and the easiest and the two most difficult contrasts are the same. This means they would be able to discriminate between both dialects similarly. On the other hand, the overlap scores predict a different scenario for the H participants. They were expected to show a bigger difference between the two dialects, as the order of difficulty varied in all positions but the easiest contrast.

When comparing the predictions between the two dialects shown in Table 6, it is also noticeable that the predicted order of difficulty varied more among BP vowels than EP vowels. In BP, the only common contrast among the two participant groups was the easiest one, /a/–/ɛ/. The order of all the other contrasts changed. The contrasts /a/–/ɛ/, /a/–/ɔ/, and /i/–/e/ had the same prediction, and the difference lay in the contrasts /e/–/ɛ/, /o/-/ɔ/, and /o/–/u/.

Table 6

Predictions for Discrimination Difficulty Based on Perceptual Overlap Scores for Each Dialect

BP	Easiest	Most difficult
H	/a/–/ɛ/ < /o/–/u/ < /o/-/ɔ/ < /i/–/e/ < /e/–/ɛ/ < /a/–/ɔ/	
ES	/a/–/ɛ/ < /e/–/ɛ/ < /o/–/u/ < /a/–/ɔ/ < /i/–/e/ < /o/-/ɔ/	
EP	**Easiest**	**Most difficult**
H	/a/–/ɛ/ < /a/–/ɔ/ < /o/-/ɔ/ < /e/–/ɛ/ < /i/–/e/ < /o/–/u/	
ES	/a/–/ɛ/ < /a/–/ɔ/ < /e/–/ɛ/ < /o/–/u/ < /i/–/e/ < /o/-/ɔ/	

Note. '<' means 'easier than' and "=" equal to.

The discrimination task was expected to support the predictions made in the categorization task, as previous studies have shown that overlap scores can predict

discrimination. Due to the differences in the perceptual overlap scores, it was anticipated that the contrasts may differ across the groups. Some contrasts were expected to be easier in one of the two dialects, as the categorization of certain vowels was more consistent in one dialect, and the overlap scores were higher.

Discrimination Results

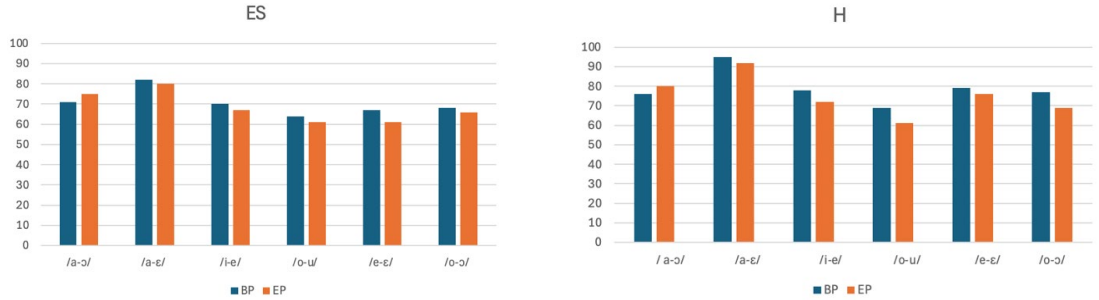
The discrimination task consisted of having participants listen to one of the six BP and EP vowels in an XAB format, with X being the target vowel. Participants were asked to identify if the first vowel (X) sounded like the first (A) or the second (B). The present study expected that difficulties in discriminating some contrasts can be explained and predicted by the categorization patterns. The order of difficulty for the discrimination of BP and EP contrasts by ES participants was expected to be similar to those of monolinguals from Elvin et al. (2018), showing that monolinguals and bilinguals have similar perception patterns. H participants were expected to show discrimination patterns more accurately than bilinguals (Flores & Rato, 2016; Rato & Carlet, 2020). Some difference between dialects in the discrimination was expected, as the dialects differ phonologically and phonetically, affecting perception (Elvin et al., 2021; Escudero & Vasiliev, 2011). Lastly, perceptual overlap was expected to predict the discrimination order (Elvin, 2016; Elvin et al., 2018; Escudero, 2009).

The results for the discrimination task showed that participants had varying success in distinguishing different BP and EP vowel contrasts, and some contrasts were more easily discriminated than others, as shown in Figure 5. The accuracy score determines the times that a contrast was correctly discriminated; the higher the accuracy score, the easier the contrast was discriminated. The contrast /a/-/ɛ/ was the easiest to distinguish for all groups and dialects, with 82% accuracy for BP contrasts and 80% for EP contrasts among ES participants, and 95% for BP and 92% for EP contrasts among H

participants. On the other hand, the /o/-/u/ contrast was the most challenging for both dialects and groups, with accuracy scores of 64% and 61% among bilinguals and 69% and 61% among heritage speakers. The EP contrast /e/-/ɛ/ was as equally to discriminate as the EP contrast /o/-/u/.

Figure 5

Discrimination Accuracy Scores on the Six BP and EP Contrasts for Bilinguals (ES) and Heritage Speakers (H)



A logistic regression generalized mixed model was conducted on Jamovi, with vowel contrasts, language background, and dialects as fixed effects and participant as random effects. The fixed effect omnibus test for contrast type yielded a highly significant result, suggesting strong accuracy differences between vowel contrasts. Post Hoc comparisons between contrast types yielded a highly significant result, suggesting strong accuracy differences between vowel contrasts. The estimates for contrast type also showed that certain contrasts were harder than others, with /a/-/ɛ/ having significantly higher accuracy, while contrast /o/-/ɔ/ had significantly lower odd of correct responses.

Notice in Table 7 that the accuracy percentages among the BP contrasts were very close, sometimes varying by one percent between some contrasts. This was the case, for example, between BP /a/-/ɔ/ and /i/-/e/ contrasts among ES, with a discrimination accuracy of 71% and 70%, or between the BP contrasts /e/-/ɛ/, /i/-/e/, /o/-/ɔ/, and /a/-/ɔ/ among H, which all varied just 1%.

Table 7

Order of Difficulty for BP and EP for Bilingual (ES) and Heritage Speakers(H) and Percentage of Discrimination Accuracy

Vowel type	Discrimination accuracy					
BP/ES	/a-ɛ/ >	/a-ɔ/ >	/i-e/ >	/o-ɔ/ >	/e-ɛ/ >	/o-u/
	82%	71%	70%	68%	67%	64%
BP/H	/a-ɛ/ >	/e-ɛ/ >	/i-e/ >	/o-ɔ/ >	/a-ɔ/ >	/o-u/
	95%	79%	78%	77%	76%	69%
EP/ES	/a-ɛ/ >	/a-ɔ/ >	/i-e/ >	/o-ɔ/ >	/e-ɛ/ =	/o-u/
	80%	75%	67%	66%	61%	61%
EP/H	/a-ɛ/ >	/a-ɔ/ >	/e-ɛ/ >	/i-e/ >	/o-ɔ/ >	/o-u/
	92%	80%	76%	72%	69%	61%

To confirm the order of difficulty presented above and check if the differences between the accuracy scores of the BP contrasts for the H participants were significant, the one-way analysis of variance (ANOVA) overall effect was run with accuracy as the dependable variable and BP contrasts as the grouping variable. The result showed a statistical significance of accuracy scores ($p < 0.001$) and a strong effect of difference ($F = 34.3$), indicating a significant difference in accuracy across the six BP contrasts. The Games-Howell post-hoc test showed that BP /a/-/ɛ/ had little variation in accuracy (with the highest mean accuracy = 0.948, the lowest standard deviation = 0.223, and standard error = 0.0112), indicating that participants in this group performed consistently well when discriminating this contrast, and this contrast was the easiest to discriminate. On the other hand, BP /o/-/u/ has the lowest mean accuracy (0.693) and the highest standard deviation (0.462), suggesting that participants in this group performed worse on average and had more variability in their performance. The other contrasts (/e/-/ɛ/, /o/-/ɔ/, /i/-/e/, /a/-/ɔ/) had moderate mean accuracies, ranging from 0.761 to 0.790, with relatively similar standard deviations and standard errors, indicating somewhat comparable performance and variability within these groups.

The same test confirmed that the highest discrimination score (/a/-/ɛ/) was significantly higher than all the others, and the second highest (/e/-/ɛ/) was significantly higher than the last hardest (/o/-/u/). However, the differences between the scores for the

other contrasts (76, 77, 78, and 79) were not significantly different from each other, signaling comparable difficulty in discriminating those contrasts. The order of discrimination difficulty for BP contrasts by Heritage speakers is (`>` means higher accuracy and '~' means not significant accuracy difference, similar result):

BP/H: /a-ɛ/ > /e-ɛ/ ~ /i-e/ ~ /o-ɔ/ ~ /a-ɔ/ ~ /o-u/

2	5	3	6	1	4
95%	79%	78%	77%	76%	69%

Running the same one-way ANOVA test for the accuracy scores by H participants with EP contrasts as a grouping variable, the same statistical significance of accuracy scores and a strong effect of difference were noticed. The order of discrimination of EP vowels by H participants was confirmed by the mean numbers of each contrast (2, 1, 5, 3, 6, 4) from the Games-Howell post-hoc test for accuracy. EP /a/-/ɛ/ stood out as the best-performing contrast with the highest mean accuracy (0.920) and the least variability (*SD* = 0.272), with a very low standard error (0.0136), suggesting that the estimate of the mean was highly precise. On the contrary, EP /o/-/u/ had the poorest performance, with the lowest mean accuracy (0.610) and the highest variability (*SD* = 0.488). The high standard error (0.0244) indicated less certainty in the mean estimate. /a/-/ɔ/ and /e/-/ɛ/ have moderately high mean accuracies (0.805 and 0.760, respectively), with moderate variability and precision, while /i/-/e/ and /o/-/ɔ/ show lower accuracy scores (0.723 and 0.685), with high variability in individual performance. The test confirmed that /a/-/ɛ/ (contrast 2) was the strongest performer, being significantly different from contrast 1. The contrast /o/-/u/ (contrast 4) had the weakest performance, and its accuracy percentage was statistically similar to /o/-/ɔ/ (contrast 6) but statically different from contrasts 3 and below. The difference in accuracy scores between the EP contrasts 1 and 5, 5 and 3, 3 and 6, and 6 and 4 were insignificant, showing relatively similar performance levels. Based on this result, the following order of discrimination for EP contrasts by H participants is

(`>` means higher accuracy and '~' means not significant accuracy difference, similar result):

EP/H: /a-ɛ/ > /a-ɔ/ ~ /e-ɛ/ ~ /i-e/ ~ /o-ɔ/ ~ /o-u/

2	1	5	3	6	4
92%	80%	76%	72%	69%	61%

Testing the accuracy results for the BP contrasts among ES participants, the one-way ANOVA test showed that the differences in accuracy among the contrasts were statistically significant. The contrast /a/-/ɛ/ outperformed the others regarding mean accuracy, while /o/-/u/ had the lowest mean accuracy. The differences in accuracy between the contrasts were statistically significant and substantial in terms of the range of mean accuracy scores (from 0.642 to 0.823). Games-Howell post-hoc test comparing the accuracy scores across the six BP contrasts confirmed the order of discrimination of BP vowels by ES participants, with /a/-/ɛ/ with the highest mean accuracy (0.823), while /o/-/u/ had the lowest mean accuracy (0.642). The relatively small standard errors suggested that the means were estimated with a good degree of precision. The contrast /a/-/ɛ/ had significantly higher accuracy ($p < 0.001$) than all other contrasts, and /a-ɔ/ was a significant difference from the most difficult contrast /o/-/u/ ($p = 0.041$). No other significances were found in most pairwise comparisons among the contrasts /i/-/e/, /o/-/u/, /e/-/ɛ/, and /o/-/ɔ/, meaning that the percentage differences among these contrasts were similar. In the end, only the BP contrast /a/-/ɛ/ differed significantly from the other contrasts, while all the other contrasts had similar difficulty. So, the order of discrimination difficulty for the BP vowels by bilingual speakers was (`>` means higher accuracy and '~' means no significant accuracy difference, similar result):

BP/ES: /a-ɛ/ > /a-ɔ/ ~ /i-e/ ~ /o-ɔ/ ~ /e-ɛ/ ~ /o-u/

2	1	3	6	5	4
82%	71%	70%	68%	67%	64%

Lastly, the one-way ANOVA test, with bilinguals' accuracy as a dependable variable and EP contrasts as a grouping variable, indicated a statistically significant difference in accuracy between at least some of the contrasts. Games-Howell post-hoc test comparing the accuracy scores across the six EP contrasts showed that the contrast /a/-/ɛ/ performed the best (mean accuracy = 0.801) and /o/-/u/ performed the worst (mean accuracy = 0.609). The order of accuracy based on the contrast means is shown below. The accuracy score for the contrast /a/-/ɛ/ was statistically similar to the contrast /a/-/ɔ/ but statistically different from all the other contrasts. The contrast /a/-/ɔ/ differed from the contrast /i/-/e/ and all the others. The contrasts /i/-/e/, /o/-/ɔ/, /e/-/ɛ/, and /o/-/u/ did not have large differences in accuracy among them. Therefore, the order of discrimination difficulty of EP contrasts by bilingual participants was ('>' means higher accuracy and '~' means not significant accuracy difference, similar result):

EP/ES: /a-ɛ/ ~ /a-ɔ/ > /i-e/ ~ /o-ɔ/ ~ /e-ɛ/ ~ /o-u/

2	1	3	6	5	4
80%	75%	67%	66%	61%	61%

The discrimination difficulty for the same dialect varied among the two participant groups (ES and H), as shown in Table 8. For the BP dialect, the easiest and the hardest contrasts were the same for both participant groups (BP /a/-/ɛ/ was the easiest to discriminate with 82% and 95% of accuracy, and the /o/-/u/ contrast was the hardest, with 64% and 69% of accuracy). Although the discrimination order for the other BP contrasts varied depending on the participant group, they were all similarly difficult to discriminate, as the percentage difference between them was not significant, as explained earlier.

Table 8

Order of Discrimination Difficulty for Bilingual and Heritage Participants Based on Statistical Analyses.

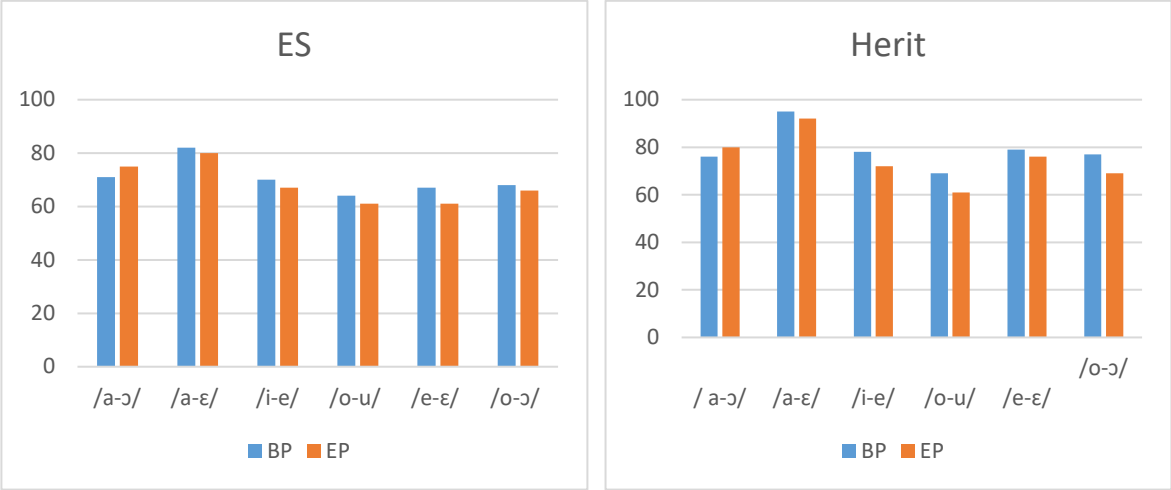
BP/ES:						EP/ES:					
/a-ɛ/ >	/ a-ɔ/ ~	/i-e/ ~	/o-ɔ/ ~	/e-ɛ/ ~	/o-u/	/a-ɛ/ ~	/a-ɔ/ >	/i-e/ ~	/o-ɔ/ ~	/e-ɛ/ ~	/o-u/
2	1	3	6	5	4	2	1	3	6	5	4
82%	71%	70%	68%	67%	64%	80%	75%	67%	66%	61%	61%
BP/H:						**EP/H:**					
/a-ɛ/ >	/e-ɛ/ ~	/i-e/ ~	/o-ɔ/ ~	/a-ɔ/ ~	/o-u/	/a-ɛ/ >	/a-ɔ/ ~	/e-ɛ/ ~	/i-e/ ~	/o-ɔ/ ~	/o-u/
2	5	3	6	1	4	2	1	5	3	6	4
95%	79%	78%	77%	76%	69%	92%	80%	76%	72%	69%	61%

Note. BP = Brazilian Portuguese; EP = European Portuguese; ES = bilingual participants; H = heritage participants.

When comparing the EP contrasts among both participant groups, the same contrasts appeared as the easiest and the most difficult ones to discriminate (EP /a/-/ɛ/ is the easiest to discriminate with 80% and 92% accuracy, and the /o/-/u/ contrast the hardest, with 61% of accuracy). For the ES participants, the EP /a/-/ɛ/ was as easy to discriminate as EP /a/-/ɔ/, but the contrast /a/-/ɔ/ was easier than /i/-/e/. The other contrasts were similarly difficult. Among H participants, EP /a/-/ɛ/ was also easier than /a/-/ɔ/, but /a/-/ɔ/ was similarly easy to /e/-/ɛ/ and /i/-/e/. The other contrasts were similarly difficult to discriminate among themselves. The discrimination accuracy percentages for BP contrasts were higher when compared to the EP contrasts, except for the /a/-/ɔ/ contrast, which had an accuracy score of 75% in EP and 71% in BP. Figure 6 shows how the BP and EP contrasts were discriminated.

Figure 6

Discrimination of BP and EP Contrasts by Bilingual (ES) and Heritage (H) Participants



The discrimination difficulty varied among the two dialects (see Figure 7). The ES participants had the same discrimination order for BP and EP (/a-ɛ/ followed by /a/-/ɔ/, /i/-/e/, /o/-/ɔ/, /e/-/ɛ/, and /o/-/u/). However, in BP, /a/-/ɛ/ was easier than /a/-/ɔ/, and /a/-/ɔ/ had a discrimination score similar to all the contrasts. In EP, /a/-/ɛ/ was similarly easy to discriminate as /a/-/ɔ/, /a/-/ɔ/ was easier than /i/-/e/, and /i/-/e/ was similarly difficult as all the other contrasts. The H participants had the easiest and the hardest BP and EP contrasts in common (/a/-/ɛ/ and /o/-/u/) but the order for the other contrasts varied.

A series of fixed effects omnibus tests from a generalized mixed model was conducted on Jamovi to evaluate how dialect, language background and contrast type influenced discrimination accuracies, with accuracy as a dependable variable, dialect, contrast, and language background as fixed factors, and participant as a random effect (Table 9). The result showed that dialect, language background, and contrast type had a statistically significant effect on the discrimination accuracy. A fixed effects parameter estimates investigated how these factors affected the discrimination accuracy. The analysis used contrast /a/-/ɔ/ as the reference category. Results indicated that ES

participants had lower odds of correct perception compared to H participants, and EP dialect also had lower odds of correct perception compared to BP dialect. The perception of contrasts varied significantly, with /a/-/ɛ/ perceived more accurately, while contrasts /i/-/e/, /o/-/ɔ/, /e/-/ɛ/, and /o/-/u/ were increasingly difficult.

Figure 7

Order of Discrimination Difficulty for BP and EP Vowels by Group

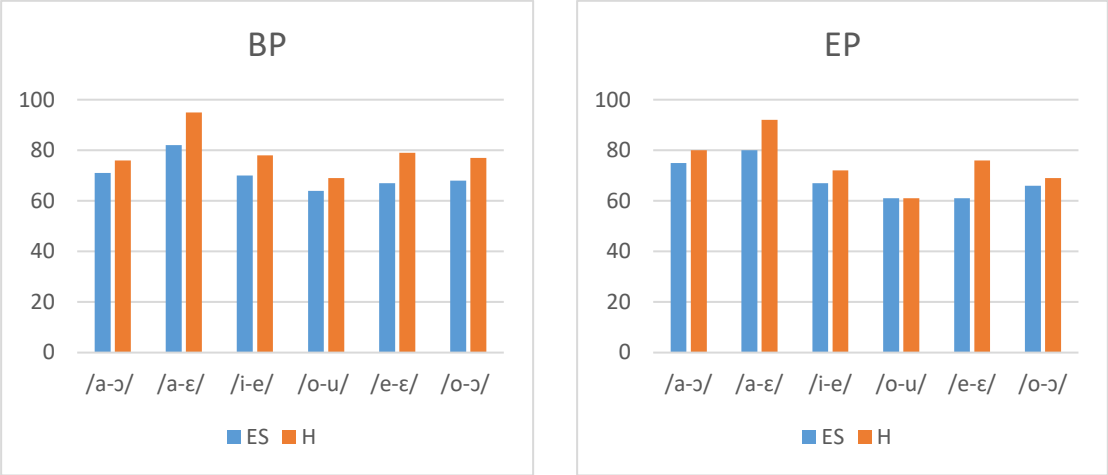


Table 9

Result from the Fixed Effects Omnibus tests on Jamovi for Factors Influencing the Discrimination Accuracies

Variable	X^2	*df*	*p*
IsBilingual	10.79	1.00	0.001
dialect	15.08	1.00	< .001
contrastnumber	297.14	5.00	< .001
IsBilingual ✻ dialect	2.91	1.00	0.088
IsBilingual ✻ contrastnumber	45.13	5.00	< .001
dialect ✻ contrastnumber	19.02	5.00	0.002
IsBilingual ✻ dialect ✻ contrastnumber	3.72	5.00	0.590

While the results for the discrimination accuracy found a main effect of contrast, dialect, and language background independently, an interaction between these factors

could show how bilingualism, dialect, and vowel contrast levels combine to influence accuracy in a non-additive way. This means, for example, that while bilingualism might lower accuracy overall, the extent of that effect could vary depending on the dialect and the difficulty of the vowel contrast. The result showed that the two-way interactions between language background and contrast type and dialect and contrast type were significant, language background and dialect was not strongly significant, and the three-way interaction between language background, dialect, and contrast type were not significant.

The significant interaction between language background and vowel contrast in the fixed effects omnibus tests indicates that the effect of being bilingual or heritage (IsBilingual) on the outcome changed significantly depending on the contrast type (contrastnumber). Post-hoc tests determined that the contrasts /a/-/ɔ/, /a/-/ɛ/, and /o/-/u/ were the most affected; the other three contrasts did not seem to have been influenced by language background.

The test also revealed a significant two-way interaction between dialect and vowel contrast, showing that the effect of dialect on accuracy varied depending on the specific contrast. Post hoc pairwise comparison was not able to show what contrasts were most affected by dialect. None of the comparisons for specific contrasts were significant, suggesting that no single contrast drove the interaction.

The fixed effects omnibus test showed no statistical significance in the interaction between language background and dialect. ES and H participants did not perform very differently based on the dialect they heard, and language background did not offer a significant advantage (or disadvantage) depending on the dialect (EP vs. BP). The lack of interaction between these two factors indicated that although dialect and language background had an effect on accuracy independently, dialect did not change the effect of language background on accuracy, and language background worked similarly across

both dialects. However, the p-value was close to 0.05, suggesting there might be a moderate influence, and with more data or a different significance threshold (e.g., 0.10), this effect could reach statistical significance. Therefore, this interaction should not be completely dismissed, as it may still be practically meaningful, and a more powerful study could yield significant results.

Lastly, no three-way interaction was found between language background, dialect, and contrast, suggesting that the effect of language background on performance across different contrasts was similar for both dialects. Each factor may have acted independently on performance, but no complex interaction affected the outcome.

Considering that one H participant reported a higher proficiency level of Portuguese in their language background questionnaire when compared to the other H participants (see Table 5, H participant n.8), this participant was excluded from the analysis, and a one-way ANOVA was conducted to check if the result from this participant would affect the significance. The result continued to show a statistically significant difference in mean accuracy among the contrasts, meaning that the participant excluded from the analysis did not affect the result.

Next, eight H participants were divided into two groups: four participants as high Portuguese self-proficiency (with an average self-proficiency rate of 2.8 and above) and four participants as low Portuguese self-proficiency (with an average self-proficiency rate of 2.4 and below). Two H participants were excluded from this analysis as they did not declare their Portuguese proficiency. This division aimed to examine whether the varied Portuguese proficiency among H participants affected the discrimination results. An independent samples t-test was conducted with accuracy as the dependent variable and proficiency as the grouping variable to assess whether there was a significant difference in accuracy between the low and high-proficiency groups. Levene's test for equality of variances was significant, indicating that there was no significant difference in accuracy

between the two groups. The effect size was extremely small, reinforcing that any observed difference between the two groups was trivial. These findings suggest no meaningful difference in accuracy between low and high proficiency groups.

In summary, there were significant differences in accuracy based on vowel contrast, with some contrasts being more challenging than others. Dialect played a role in discrimination, with EP being more difficult to discriminate than BP. Group differences were also evident, with H participants demonstrating higher accuracy than bilinguals, regardless of the contact with their HL. Furthermore, important interactions were observed, particularly between the participant group and vowel contrast, as well as between dialect and vowel contrast. This indicates that the effects of dialect and group on accuracy were influenced by the specific vowel contrasts being tested. This complex interaction reveals how language background, dialect, and phonological difficulty interact in the context of speech discrimination tasks.

In the next section, the discrimination results for both Portuguese dialects will be evaluated and compared to the predictions made by the perceptual assimilation overlap score.

Perceptual Overlap and Discrimination

Recall that perceptual overlap was expected to predict accuracy in the discrimination task. The contrasts with the higher overlapping scores were expected to have lower discrimination accuracy than those with lower perceptual scores.

The overlap scores and discrimination accuracy for the ES participants for each Portuguese contrast are shown in Table 10 below, ordered from the lowest to the highest overlap score, and Figure 8 shows the order of predicted discrimination difficulty compared to the order of discrimination accuracy. The perceptual overlap scores correctly predicted the BP contrast /a/-/ɛ/ as the easiest, with overlap scores of 13 and

discrimination accuracy percentages of 82%. The contrast /o/-/ɔ/ was predicted to be the hardest one, with overlap scores of 58, but in the discrimination, this contrast did not have the lowest discrimination accuracy score (its discrimination accuracy was 68%, in front of /e/-/ɛ/ and /o/-/u/). However, the one-way ANOVA test found that the discrimination accuracy percentages between the contrasts /o/-/ɔ/ (68%) and /o/-/u/ (64%) were similar. This means that the overlap scores were not completely wrong. The EP contrast /a/-/ɛ/ was correctly predicted as the easiest one, with overlap scores of 0 and discrimination accuracy percentages of 80%, followed by the contrast /a/-/ɔ/ as the second easiest, with overlap scores of 25 and accuracy percentages of 75%. Just as it happened to BP, the contrast /o/-/ɔ/ was not correctly predicted as the most difficult contrast to discriminate, but the statistical analysis showed that the discrimination difference between /o/-/u/ and /o/-/ɔ/ was not significant.

Table 10

Perceptual Overlap Scores and Discrimination Percentages Correct for Each Portuguese Vowel Contrast for the Bilingual (ES) Participants

BP	**/a/-/ɛ/**	**/e/-/ɛ/**	**/o/-/u/**	**/a/-/ɔ/**	**/i/-/e/**	**/o/-/ɔ/**
Overlap score	13	36	38	41	47	58
Discrimination accuracy	82	67	64	71	70	68
EP	**/a/-/ɛ/**	**/a/-/ɔ/**	**/e/-/ɛ/**	**/o/-/u/**	**/i/-/e/**	**/o/-/ɔ/**
Overlap score	0	25	36	40	52	69
Discrimination accuracy	80	75	61	61	67	66

Note. The Order of the Vowel Pairs is from Lowest to Highest Overlap Score for Each Language.

Figure 8

Order of Perceptual Overlap (PO) versus Discrimination (DI) Result – Bilingual (ES) Participants

BP/ES:

PO: /a–ɛ/ > /e–ɛ/ > /o-u/ > /a-ɔ/ > /i-e/ > /o-ɔ/

DI: /a-ɛ/ > /a-ɔ/ ~ /i-e/ ~ /o-ɔ/ ~ /e-ɛ/ ~ /o-u/

EP/ES:

PO: /a-ɛ/ > /a-ɔ/ > /e-ɛ/ > /o-u/ > /i-e/ > /o-ɔ/

DI: /a-ɛ/ ~ /a-ɔ/ > /i-e/ ~ /o-ɔ/ ~ /e-ɛ/ ~ /o-u/

For the H participants (see Table 11 & Figure 9), the only BP contrast correctly predicted by the perceptual overlap was /a/-/ɛ/, with overlap scores of 10 and discrimination accuracy percentages of 95%. Interestingly, the BP contrast /o/-/u/ was predicted as the second easiest one, but in the discrimination task, it was the most difficult contrast among all dialects and language groups. The overlap scores correctly predicted the EP contrasts /a/-/ɛ/ and /a/-/ɔ/ as the two easiest discriminating contrasts, with overlap scores of 0 and 27 and discrimination accuracy percentages of 92 and 80%, and the contrast /o/-/u/ as the hardest one, with overlap scores of 59 and discrimination accuracy percentages of 61%, respectively.

Table 11

Perceptual Overlap Scores and Percentages Correct for Each Portuguese Vowel Contrast for the Heritage (H) Participants.

BP	**/a/-/ɛ/**	**/o/-/u/**	**/o/-/ɔ/**	**/i/-/e/**	**/e/-/ɛ/**	**/a/-/ɔ/**
Overlap score	**10**	17	22	32	36	39
Discrimination accuracy	95	69	77	78	79	76
EP	**/a/-/ɛ/**	**/a/-/ɔ/**	**/o/-/ɔ/**	**/e/-/ɛ/**	**/i/-/e/**	**/o/-/u/**
Overlap score	0	27	30	44	55	59
Discrimination accuracy	92	80	69	76	72	61

Note. The Order of the Vowel Pairs is from Lowest to Highest Overlap Score for Each Language.

Figure 9

Order of Perceptual Overlap versus Discrimination Result – Heritage (H) Participants

BP/H:

PO: /a–ɛ/ > /o–u/ > /o-ɔ/ > /i–e/ > /e–ɛ/ > /a–ɔ/

DI: /a-ɛ/ > /e-ɛ/ ~ /i-e/ ~ /o-ɔ/ ~ /a-ɔ/ ~ /o-u/

EP/H:

PO: /a–ɛ/ > /a–ɔ/ > /o-ɔ/ > /e–ɛ/ > /i–e/ > /o–u/

DI: /a-ɛ/ > /a-ɔ/ ~ /e-ɛ/ ~ /i-e/ ~ /o-ɔ/ ~ /o-u/

A simple linear regression test was run on Jamovi to check for a correlation between the overlap score and the discrimination accuracy. Overall, overlap and discrimination had a significant effect, meaning that perceptual overlap could predict discrimination accuracy. The results showed that as perceptual overlap increases, discrimination performance decreases, suggesting that perceptual overlap has a meaningful influence on discrimination performance. The intercept was highly significant, meaning that the model's baseline prediction was reliable.

Thus, although the perceptual overlap scores did not present the same order of discrimination difficulty, statistical analysis confirmed that discrimination performance could indeed be explained by the perceptual overlap score, as demonstrated in other studies (Elvin et al., 2018; Levy, 2009; Vasiliev, 2013).

CHAPTER 5: DISCUSSION AND CONCLUSION

This study explored how bilingual ES and monolingual heritage university students learning Portuguese as a Second Language categorized seven Portuguese vowels from two dialects, BP and EP, and distinguished six minimal pairs from both dialects. It sought to determine whether there were differences in categorization scores between the BP and EP dialects and between bilingual and heritage groups, as well as differences in discrimination scores influenced by dialects and language background. Additionally, it aimed to investigate whether perceptual overlap scores could predict discrimination difficulties.

Although participants categorized the BP and EP vowels into similar CE categories (except for a few vowels), BP vowels had higher classification percentages for most vowels, indicating that listeners were more certain when classifying BP vowels than EP vowels. The classification order changed according to the dialect, showing a possible influence of dialect. Lastly, H participants categorized the Portuguese vowels more accurately, with less variation in categorization percentages, confirming the hypothesis that language background influences categorization and that early contact with the language is an advantage to perception.

The results from the discrimination task were consistent with those from the categorization task, demonstrating the influence of dialect and language background. Just like in the categorization task, ES and H participants performed better in BP than EP. Language background also played a role, with H participants discriminating the Portuguese contrasts more accurately than ES, showing that early exposure to a HL is an advantage in discrimination. Furthermore, the results from the discrimination confirmed that perception overlap scores can predict discrimination difficulties.

The fact that the H participants distinguished BP more accurately than EP was unexpected, as their HL is EP. In the language background questionnaire (see Appendix C), the H participants mainly identified themselves with Azorean Portuguese culture, and half of them spoke European Portuguese with their parents and/or grandparents at home. This indicates that H participants were somehow familiar with their HL, even though they were not fluent, and therefore, it could imply better discrimination toward their heritage dialect. However, the preference for BP vowels may be due to the acoustic properties of the BP and EP vowels. For example, BP has longer stressed vowels and a more significant intrinsic pitch effect than EP, which could make BP vowels easier to discriminate (Escudero et al., 2009; Alispahic et al., 2017).

The discrimination order of BP vowels by ES participants was consistent with the previous study by Elvin et al. (2018), as shown in Table 12. This comparison shows that, despite some variation, the easiest and the hardest contrasts remained consistent between studies, namely /a/-/ɛ/, /a/-/ɔ/, and /o/-/u/. Although the other contrasts do not follow the same order, the accuracy percentages were not too different. For example, the BP /i/-/e/ contrast had 72% accuracy in Elvin and 70% in the present study. Elvin's BP contrasts /o/-/ɔ/ and /e/-/ɛ/ had very close accuracy scores (73.5% & 72%), just like the present study (68% & 67%). Changing to EP, the comparison is similar. In both Elvin et al. (2018) and the present study (see Table 13), EP /a/-/ɛ/ and /a/-/ɔ/ were the easiest EP contrasts to discriminate, and the EP contrast /o/-/u/ is the most difficult to discriminate by bilinguals. Although the order of the other contrasts varies, the discrimination accuracy percentages do not seem to vary significantly among them. It is also important to consider that statistical analysis demonstrated the discrimination difficulty among the five most difficult contrasts, hinting that the order of discrimination difficulty does not mean much, as all five contrasts have similar percentages.

Table 12

Order of Discrimination of BP Vowels by Bilingual Participants as Elvin et al. (2018), and the Present Study

Elvin et al. (2018): BP dialect by bilingual participants											
BP	/a-ɛ/	>	/a-ɔ/	=	/o-ɔ/	>	/e-ɛ/	=	/i-e/	>	/o-u/
Bilinguals	89%		73.5%		73.5%		72%		72%		68%
Present study: BP dialect by bilingual participants											
BP	/a-ɛ/	>	/a-ɔ/	~	/i-e/	~	/o-ɔ/	~	/e-ɛ/	~	/o-u/
Bilinguals	82%		71%		70%		68%		67%		64%

Table 13

Order of Discrimination of EP Vowels by Bilingual Participants as Elvin et al. (2018), and the Present Study

Elvin et al. (2018): EP dialect by bilingual participants											
EP	/a-ɛ/	>	/a-ɔ/	>	/o-ɔ/	~	/e-ɛ/	>	/i-e/	>	/o-u/
Bilinguals	87.5%		81%		74.5%		73%		65%		62%
Present study: EP dialect by bilingual participants											
EP	/a-ɛ/	~	/a-ɔ/	>	/i-e/	~	/o-ɔ/	~	/e-ɛ/	~	/o-u/
Bilinguals	80%		75%		67%		66%		61%		61%

Remember that Elvin et al. (2018) found that bilinguals performed similarly to monolinguals in discriminating Portuguese vowels, indicating that experience with similar languages did not significantly affect the perception of non-native contrasts. When comparing the results of BP vowels from ES participants in this study with those of monolinguals from that study and Vasiliev (2013), as shown in Table 14, the discrimination order in the three studies is also similar. In the three studies, the easiest BP contrast to discriminate is /a/-/ɛ/. In Elvin et al. (2018) and the present study, the hardest contrast to discriminate was /o/-/u/, with 62% and 64% accuracy, while in Vasiliev (2013), the lowest discrimination score was for the BP contrast /i/-/e/, with 68.5%. However, statistical analysis in the present study showed that the percentages between the contrasts /i/-/e/ and /o/-/u/ were similar. For the EP vowels, the two easiest contrasts (/a/-/ɛ/ and /a/-/ɔ/) and the hardest one (/o/-/u/) were the same among monolinguals from Elvin et al. (2018) and bilinguals from the present study. Just like in BP, although the

order varied between the two studies, the percentages do not vary much. For example, in Elvin et al. (2018), /i/-/e/ had an accuracy of 65% whereas in the present, it had 67%; Future statistical analysis could tell if these numbers are statistically similar, but it seems that they are.

Table 14

Order of Discrimination of BP Vowels by Monolingual Participants and Bilingual Participants as in Vasiliev (2013), Elvin et al. (2018), and the Present Study.

Vasiliev (2013): BP dialect by monolingual participants													
BP Monolinguals	/a-ɛ/ 99%	>	/e-ɛ/ 95%	~	/o-ɔ/ 94%	>	/a-ɔ/ 82.5%	>	/i-e/ 68.5 %	;	/o-u/ 74.5 %	~	/i-e/ 68.5 %
								~	/o-u/ 74.5 %				
Elvin et al. (2018): BP dialect by monolingual participants													
BP Monolinguals	/a-ɛ/ 87%	>	/o-ɔ/ 77.5%	>	/e- ɛ/ 76%	>	/a –ɔ/ 73%	>	/i -e/ 67%	>	/o-u/ 62%		
Present study: BP dialect by bilingual participants													
BP Bilinguals	/a-ɛ/ 82%	>	/a-ɔ/ 71%	~	/i-e/ 70%	~	/o-ɔ/ 68%	~	/e-ɛ/ 67%	~	/o-u/ 64%		
Elvin et al. (2018): EP dialect by monolingual participants													
EP Monolinguals	/a–ɛ/ 87%	>	/a–ɔ/ 76%	>	/o-ɔ/ 74%	~	/e-ɛ/ 73%	>	/i-e/ 65%	>	/o-u/ 56%		
Present study: EP dialect by bilingual participants													
EP Bilinguals	/a-ɛ/ 80%	~	/a-ɔ/ 75%	>	/i-e/ 67%	~	/o-ɔ/ 66%	~	/e-ɛ/ 61%	~	/o-u/ 61%		

Therefore, it seems that bilinguals in the present study performed similarly to monolinguals in the other two studies, confirming previous findings that monolinguals and bilinguals discriminate BP vowels similarly. However, it is also important to consider that the stimuli were very short and quick, making it hard to identify the acoustic properties of each phoneme properly and disregarding some cognitive functions that may help discrimination at the lexical level, as Elvin et al. (2018) suggested.

While bilingualism does not appear to enhance Portuguese vowel perception, early contact with a HL improves discrimination accuracy (Flores & Rato, 2016; Osborne, 2021; Rato & Carlet, 2020). Consistent with prior research, H participants demonstrated better perception of the target vowels compared to both monolinguals and bilinguals, regardless of HL proficiency. This finding emphasizes the importance of maintaining the use of HL within families, as early exposure can enhance phonetic perception, which in turn benefits formal language learning later in life (Choi et al., 2017; Osborne, 2021; Zhou et al., 2021).

To address potential perceptual challenges faced by learners, educators should prioritize specific phonetic training that considers the influence of the students' native language (Elvin et al., 2020; Tuninetti et al., 2020; Wanrooij et al., 2013). Targeted instruction focusing on problematic phonetic contrasts can lead to more effective learning outcomes. As suggested by Diaz Granado (2011), teachers should also carefully monitor students' pronunciation, as mispronunciation often indicates difficulties in sound perception. After a short period of time, perceptual training can not only improve learners' performance in the identification of the target vowels, but it can also help the retention of the sound for two months after training, showing that L2 adult learners' perceptual system could change with appropriate training (Rato, 2014; Wang et al., 2022).

Instructional materials for Portuguese as a Second Language (PSL) educators could address specific challenges, such as the difficult Portuguese contrasts /o/-/u/ and /e/-/ɛ/ and their complex allophonic variations, by offering targeted phonetic exercises. In the contrast /o/-/u/, for example, the phoneme /o/ has three different allophones in Portuguese: [ʊ], [o], and [ɔ], which account for the contrasts /o/-/u/ and /o/-/ɔ/ among the most difficult contrasts to discriminate (Zampaulo, 2019). While the allophone [o] is the most common, in BP, the allophone [ʊ] usually occurs at the end of words following the

stressed syllables, such as in tudo (ˈt[u]d[ʊ]) and saco (ˈsak[ʊ]). [ɔ] usually occurs in stressed syllables when there is a morphological difference between words, producing minimal pairs such as avˈo/ ‘grandfather’ and avˈɔ/ ‘grandmother’, and jogo ‘game’ noun (ˈʒ[o]gʊ) and jogo ‘to play’ verb (ˈʒ[ɔ]gʊ). In EP, /o/ sounds like [u] before or after the stressed syllables, such as sapo ‘frog’ (ˈsap[u]) and português ‘Portuguese’ (p[u]ɾtuˈgeʃ). The mid-vowel contrast /e/-/ɛ/ is found in minimal pairs such as sˈe/de ‘thirst’ – sˈɛ/de “headquarters.” However, these contrasts are considered more marked as they have a low functional load, meaning they are not frequent in BP (Osborne, 2021). In context, though, it seems difficult to understand how these two contrasts and other contrasts such as /a/-/ɔ/ and /i/-/e/ could be confused, showing that context plays a role in discrimination. Both vowels in these contrasts might be very distinguishable in context, as one is not an allophone of the other. Therefore, context is also a differentiator for the discrimination of these two vowels.

Future research should include acoustic analysis of Azorean Portuguese, offering a comparative perspective with other dialects. Additionally, detailed acoustic comparisons of participants’ native production data could provide a more accurate prediction of their non-native categorization and discrimination difficulties (Elvin et al., 2021). The effect of dialects could also be checked by comparing results from a discrimination task taken by native EP speakers from the Azores and learners of EP. Furthermore, the differences among BP dialects should be examined to better understand second language learners' challenges when perceiving different varieties from the same country. Future studies could investigate how native speakers of BP perceive EP dialects and vice-versa to determine if dialectal variation interferes with the perception of different native dialects. Lastly, a discrimination task at a word level would confirm the importance of context for discriminating certain contrasts.

It would also be valuable to compare learners' performance at varying levels of L2 acquisition (e.g., elementary vs. advanced), as previous research has shown that exposure to L2 affects perception, word recognition, and production (Kendall, 2004; Wang et al., 2022). Factors such as age, age-of-learning (AOL), and Portuguese proficiency level should also be considered in future investigations.

The current study did not examine the influence of Spanish vowels on the perception of Portuguese as a third language (L3). Although it is important to acknowledge the influence of a second language on production, language processing, and cognition (e.g., Elvin, 2024; Grijalva et al., 2013; Kroll & Bialystok, 2013), the goal of the present study was to confirm previous findings that bilinguals perform similarly to monolinguals, regardless of the L2 language, and contact with a HL positively impacts the HL's perception. The present study confirmed that bilinguals performed similarly to monolinguals when perceiving Portuguese vowels.

Regarding the methodology, some studies (Elvin et al., 2021; Faris et al., 2018; Vasiliev, 2013) suggest that the task order in the perception experience can influence the results. Because of this, participants should take the categorization task first, followed by the discrimination task in the same session, as exposure to the target sounds in the discrimination task can influence the categorization task's result. Studies analyzing this difference could help understand if this factor significantly influences discrimination.

Additionally, traditional discrimination methods, such as the two-interval forced-choice task (2IFC), may introduce biases, as suggested by Schouten et al. (2003). They claim that the two-interval forced-choice task (2IFC) encourages listeners to rely on subjective criteria, which may resemble categorical judgment rather than true phonetic discrimination. To achieve more reliable phoneme discrimination, especially for vowels, the discrimination task should involve more natural stimuli, replicating the continuous acoustic variation and leaving less evidence of categorical boundaries.

Similarly, Wang et al. (2022) pointed out that vowels in isolation do not take into consideration the influence that phonetic-phonological context can have on the target vowel (see Escudero et al., 2022; Levy, 2009; Strange et al., 2007). In Smirnova Henriques et al. (2019), participants' comments illustrate the choice of using nonce words. Native Russian speakers took a perception task of BP vowels using real words, and they emphasized that, in real life, the words were easier to discriminate as they were in a context, while in the test, just with audio, they were much harder. One of them humorously remarked that "In real life, we do not have difficulties in differentiating *avô* and *avó*. Brazilians always say *meu avô* or *minha avó*, but these crazy people differentiate the words in some way" (Smirnova Henriques et al., 2019, p. 75). These comments support the idea that perception is influenced by lexical context, and future studies could compare vowels in isolation and in context.

Although most studies focus on learner groups rather than individuals, drawing general conclusions about performance tasks may not account for individual differences in the realization of vowels (Elvin et al., 2021). This could happen because not only extra-linguistic factors (such as age of acquisition, length of residence, language use, and motivation) but also cognitive differences among individuals can affect L2 acquisition. Therefore, the acoustic comparison of L1 categories and the categories of the target language at the individual level (using individual overlap scores) could be a better predictor of perceptual performance. Furthermore, the use of nonce words in the perception study could reflect the reality learners face more closely when learning a new language, as speech is not learned as isolated sounds but within words.

Lastly, future work should also expand the differences between online and in-person perception tasks. A recent surge in studies has explored this area, as online tasks are more flexible and enable the recruitment of more participants. However, they present challenges such as reduced control over environmental factors (like background noise)

and participant engagement, which can affect the reliability of results (Mathôt & March, 2022).

In conclusion, the present study found that contact with a HL helps identify the sounds in the HL, regardless of proficiency level in the HL: It also presented which sounds are easier and more difficult to discriminate, with the hope that this information help educators. However, some limitations can guide future research to the analyze the acoustic properties of Azorean Portuguese dialect and its categorization and discrimination. Additionally, critical steps should refine the methodologies used in discrimination tasks and consider individual and cognitive factors in L2 acquisition. Finally, the growing use of online perception tasks presents new opportunities, but also challenges that must be addressed in future research.

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APPENDICES

APPENDIX A: CONSENT FORM

Consent Form

You are invited to participate in a study conducted by [the graduate student Maria Eliza Amadeo. We hope to learn how Portuguese vowels are perceived by learners of Portuguese and Heritage speakers of Portuguese from California. You were selected as a possible participant in this study because you are a Portuguese learner and/or a Heritage speaker of Portuguese.

Your researcher will inform you of the instructions, and you will be asked to simply listen to the stimuli and give your responses either by clicking on a mouse or pressing a button on a keyboard. You will also be asked to fill out a language background form.

The study will not benefit you directly but will tell us more about how people learn new words in a second language. All aspects of the study, including results, will be confidential, and only the researchers will have access to information on participants. The group results may be published in scientific journal articles. No individual results will be reported.

If you are a Portuguese learner at Fresno State, you will be offered extra credits for your participation. You will have no expenses to participate in this experiment.
If you are a community member, you are expected to pay for your transportation to go to campus.

Your decision whether or not to participate will not prejudice your future relations with California State University, Fresno. If you decide to participate, you are free to withdraw your consent and to discontinue participation at any time without penalty. The Committee for the Protection of Human Subjects at California State University, Fresno has reviewed and approved the present research.

If you have any questions, please ask us. If you have any additional questions later, Dr. Jaydene Elvin (559) 278-3056 will be happy to answer them. Questions regarding the rights of research subjects may be directed to Dr. Jennifer Randles, Chair, CSU Fresno Committee on the Protection of Human Subjects, (559) 278-2448.

You will be given a copy of this form to keep.

YOU ARE MAKING A DECISION WHETHER OR NOT TO PARTICIPATE. YOUR SIGNATURE INDICATES THAT YOU HAVE DECIDED TO PARTICIPATE, HAVING READ THE INFORMATION PROVIDED ABOVE.

Printed name _____ **Date** _____

Signature _____

APPENDIX B: BACKGROUND QUESTIONNAIRE

1. How old are you? _____
2. What's your gender? ___ male ___female ___ prefer not to declare
3. What is your level of education?

___ Graduate School (Doctorate)
___ Graduate School (Master)
___ College (Bachelor)
___ High School
___ Middle School
___ Elementary School
___ Other - specify: _____

4. Do/did you have a hearing impairment or reading difficulties (e.g., difficulties learning to read), or language development or speaking difficulties (e.g., delayed language development, stuttering, lisping, etc.)?

____ yes ____ no

4.1 If yes, check any/all that apply:

____ Hearing
____ Reading - difficulties learning to read, etc
____ Speaking - stuttering, lisping, etc
____ Delayed Language Development

5 What is your native language?

(Native language is the language you have spoken from birth or since you are a child; more than one option is possible)

___ English
___ Spanish
___ Azorean Portuguese
___ European Portuguese
___ Brazilian Portuguese
___ Other - specify: _____

5.1. If you chose more than one native language, which language are you more fluent in overall?

(e.g. equally fluent in English and Spanish; English)
___ Equally fluent
___ More fluent in _____

5.2. If you have more than one native language, when did you start speaking those languages?

E.g.: English / from birth; Portuguese / 5 yo

Language	Time

5.3. If you chose more than one native language, how often do you use them during the day?

E.g.: English / 3

Language	How often (1= never; 5= always)
	___ 1 ___ 2 ___ 3 ___4 ____ 5
	___ 1 ___ 2 ___ 3 ___4 ____ 5

6. Which cultures/languages do you identify with more strongly? More than one option is possible.

___ English
___ Spanish
___ Azorean Portuguese
___ European Portuguese
___ Brazilian Portuguese
___ Other - specify: _____

7. Overall, you consider yourself... (only one option is possible)

Monolingual - speak only one language fluently
Bilingual - speak two languages fluently
Multilingual - speak more two three languages fluently
Other - specify: _____

8. Please tell us every place you have lived for at least three months or more, starting with your infancy and childhood. Country, City or Region, Dates lived there, Between which ages?

Country	City/Town/Region	Between which ages?
e.g.: US, California	California, Fresno	Birth-present
Brazil	Rio de Janeiro	0-39

9. What is your mother's native language?

___ English
___ Spanish
___ Azorean Portuguese
___ European Portuguese
___ Brazilian Portuguese
___ Other - specify: ____

9.1. What's her place of birth, age of arrival in the US, and current residence?

Place of birth	Age of arrival in the US	Current residence

10. What is your father’s native language?

___ English
___ Spanish
___ Azorean Portuguese
___ European Portuguese
___ Brazilian Portuguese
___ Other - specify: _____

10.1. What's his place of birth, age of arrival in the US, and current residence?

Place of birth	Age of arrival in the US

11. What are your grandparents' native languages?

Grandmothers		Grandfathers	
___ English ___Spanish ___ Azorean Portuguese ___ European Portuguese ___ Brazilian Portuguese ___ Other - specify:	___ English ___Spanish ___ Azorean Portuguese ___ European Portuguese ___ Brazilian Portuguese ___ Other - specify:	___ English ___Spanish ___ Azorean Portuguese ___ European Portuguese ___ Brazilian Portuguese ___ Other - specify:	___ English ___Spanish ___ Azorean Portuguese ___ European Portuguese ___ Brazilian Portuguese ___ Other - specify:

11.1. Where did your grandparents grow up?

Place of birth - town/region/country	Age of arrival in the US
E.g.: grandmother - Rio de Janeiro, Brazil grandfather - Rio de Janeiro, Brazil	US at 45 yo US at 50 yo

12. When you were a child, what language did you speak <u>at home</u> with your family/friends?

E.g. : grandparents / Brazilian Portuguese / 5; parents / Brazilian Portuguese / 5; cousins / Brazilian Portuguese / 4; friends / English / 4

Person	Language	How often - **1= never; 5= always**
		___ 1 ___ 2 ___ 3 ___4 ____ 5
		___ 1 ___ 2 ___ 3 ___4 ____ 5

13. Was there any other adult from a different country, or from a different city/region, who spent a <u>large amount</u> of time with you when you were growing up (a primary caregiver such as an aunt or a live-in housekeeper)?

If so, who, where they were born, what language they spoke with you, and during what ages in your childhood?

E.g.: Aunt / Rio de Janeiro, Brazil / Brazilian Portuguese / 03-15 yo

14. What language did you speak in school? How old were you? i.e. language of instruction.

If language immersion, please specify in comments below which grades and number of hours/week Comments: _____

15. What other language(s) have you studied or learned?

___ English
___ Spanish
___ Azorean Portuguese
___ European Portuguese
___ Brazilian Portuguese
___ Other - specify: _____

15.1. Please tell us how long you have spoken them, and how well you speak and understand them. e.g. Spanish/ 2006-2008/ high school, 2, 3

(1= hardly at all; 5= highly fluent)

Language	Years used	Where did you use it?	How well do you speak it?	how well do you understand it?

16. Please evaluate your current level of proficiency for all languages/accents that you have ever been exposed to, considering the following skills:

(1= beginner; 5= very advanced)

	Language:	Language:	Language:	Language:
Oral comprehension				
Oral production				
Writing proficiency				
Reading proficiency				
Pronunciation				

17. Have you ever taken Portuguese classes at Fresno State?

___ Yes - mark all that apply: ___ No
___ Port 1A
___ Port 1B
___ Port 2A
___ Port 2B
___ Portuguese Culture

18. Please provide your e-mail address if you are willing to be contacted for further inquiries related to this study in the future.

THANK YOU.

APPENDIX C: SELF-DECLARED PROFICIENCY

Self - Declared proficiency in Portuguese (BP or EP) by Heritage Participants.
(1 = beginner; 5 = very advanced)

part	Oral comprehension	production	writing	reading	pronunciation	average
1	2	2	3	2	1	2
2	3	2	3	3	3	2.8
3	3	3	1	2	3	2.4
4	4	2	2	3	3	2.8
5	nd	nd	nd	nd	nd	nd
6	4	2	2	3	3	2.8
7	1	1	1	1	1	1
8	2	3	3	3	4	3
9	3	2	2	2	2	2.2
10	nd	nd	nd	nd	nd	nd

Self-Declared Proficiency in Spanish by Bilingual Participants.
(1 = beginner; 5 = very advanced)

part	Oral comprehension	production	writing	reading	pronunciation	average
1	4	4	3	4	4	3.8
2	5	5	5	5	5	5
3	4	3	2	2	4	3
4	5	4	3	5	5	4.4
5	5	5	4	4	4	4.4
6	5	4	1	3	4	3.4
7	5	5	4	5	4	4.6
8	5	3	3	4	3	3.6
9	5	5	5	5	5	5
10	5	5	4	4	5	4.6
11	5	5	4	5	5	4.8
12	5	4	4	4	5	4.4
13	5	5	5	5	4	4.8

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